The niche of Romik's ambidextrous sofa in convex-linear data, and a stability bound under a curvature condition

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Abstract

Baek proved that Gerver's sofa is the maximum-area moving sofa for a corridor with one corner, by writing a sofa as a convex cap minus a niche and obtaining optimality from the concavity of a quadratic upper bound. We carry that architecture to the ambidextrous moving sofa problem, which is open, and to Romik's candidate Σ of area $A_R^* = 1.6449552184\ldots$

The niche of an ambidextrous sofa is expressed exactly in convex-linear data: writing $H(\theta) = h_K(\mu_\theta)$ for the cap's support function, the two arms α_1, α_2 and the reach σ are affine in H, and $|N| = \int_0^{\pi/2}[\frac{1}{2}(\alpha_2^+)^2 + \frac{1}{2}(\sigma - \alpha_1)^2 - \frac{1}{2}(\alpha_1^-)^2]\,dt$ whenever the associated sweep is injective. Injectivity follows from a single linear condition (RC), that the cap's radius of curvature be at most the corridor width; the proof reduces each of the three required statements to a forced harmonic oscillator whose kernel is non-negative because the rotation angle is $\pi/2$. (RC) is classical: by the dual of Blaschke's rolling theorem it says the cap slides freely inside a unit ball, up to the corridor facets. What is new is its use.

For the functional $Q = |C_2| - 2V$ we prove $Q(\Sigma) = A_R^*$; that $\delta Q(\Sigma) = 0$, so Romik's ODE families are exactly the critical-point equations of Q, identically in his constants; and that $\delta^2 Q < 0$, with an explicit constant. Two Cauchy–Schwarz steps decouple the two halves of $[0, \pi]$ and reduce the second variation to a pair of Sturm–Liouville eigenvalue problems, certified from below by Sturm oscillation; this gives $\frac{1}{2}\delta^2 Q \le -\frac{2}{3}\|\eta\|^2_{L^2(0,\pi)}$. Not splitting at all turns the second variation into a single 2×2 system whose first eigenvalue is the sharp constant $c^* = 0.7309566\ldots$, certified to exceed $\frac{73}{100}$. The same system, with the cell boundaries as parameters, proves concavity for every cell whose sign sets are ordered and anchored, by a one-line monotonicity reduction to a one-parameter family certified by an adaptive interval covering; on the mirror class the constant is uniform, $\frac{3}{10}$, and the resulting stability bound assumes no smallness of $H - H_\Sigma$ beyond (RC). On the convex domain $\mathcal{D}$ cut out by (RC), a separation inequality and Σ's own sign pattern,

$$|T| \;\le\; A_R^* - \tfrac{73}{100}\,\|H - H_\Sigma\|^2_{L^2(0,\pi)}$$

for every ambidextrous sofa T with cap data in $\mathcal{D}$, and Σ is the unique member attaining A_R^*. The domain is a C^2-neighbourhood of Σ: its radius along $\sin(2k\theta)$ scales like k^{-2}.

Two limits are recorded rather than smoothed over. The separation inequality is independent of (RC) and cannot be removed. And V is the exact flux of the advancing wedge boundary, so $V \ge |N|$ by Reynolds' transport theorem and Q bounds nothing where the sweep fails to be injective; in particular the construction does not recover Gerver's sofa, whose cap violates (RC). Separately, a two-hallway relaxation gives the first upper bound specific to the ambidextrous problem: the area of a left- and a right-turning hallway at angle $\pi/4$ is bounded by translates of one explicit profile integral, and a four-case analysis of the corner offsets removes every hypothesis, giving

$$A_{\mathrm{ambi}} \;\le\; 2\sqrt{2} - 1 \;=\; 1.8284271\ldots$$

unconditionally. The bound is attained by an explicit placement, so it is exactly the value of the two-hallway relaxation; it improves on the 2.2195 inherited from the one-corner theorem, and with Romik's sofa it brackets the ambidextrous supremum within a factor of 1.1116. Optimality of Σ remains open.

1 Introduction

The moving sofa problem asks for the largest area of a connected shape that can be manoeuvred around a right-angled corner of a hallway of unit width. It was settled by Baek [1], who proved Gerver's sofa [2] optimal. Romik [5] posed the *ambidextrous* variant, in which the shape must turn corners of both handedness, exhibited a candidate Σ of area

$$A_R^* = 1.6449552184254408\ldots,$$

and asked whether it is optimal. That question is open, and this note does not settle it. What follows is a structural analysis of the ambidextrous problem in the coordinates in which the constraint data is affine, and a local optimality statement for Σ on an explicitly described domain.

The coordinates

Every ambidextrous sofa is $C_2 \setminus (U \sqcup \rho U)$ for a convex *cap* C_2 and a *niche* U swept by the hallway's inner corner, with ρ the reflection in $y = \frac{1}{2}$. Writing $H(\theta) = h_{C_2}(\mu_\theta)$ for the cap's support function on $[0, \pi]$, the corner path is *affine* in H (Lemma 18), and so are the two arm lengths α_1, α_2 and the reach σ that determine the sweep. In these coordinates the cap area is an exact quadratic form,

$$|C_2| = \int_0^\pi (H^2 - H'^2)\, d\theta - H(0) - H(\pi)$$

(Proposition 24), and the niche area has the closed form (20). The functional $Q := |C_2| - 2V$, with V the flux of the advancing sweep, is therefore an explicit quadratic polynomial in H on each cell of the sign pattern of (α_1, α_2).

The results

The analysis rests on one geometric hypothesis, that the cap is nowhere flatter than a circle of the corridor's width:

$$\text{(RC)} \qquad (H + H'')_{\mathrm{ac}} \le 1,$$

which is linear in H and hence convex. It is classical, being the sliding-ball condition dual to Blaschke's rolling theorem (Remark 39); what is new is its use. Theorem 40 shows that (RC) implies all three injectivity conditions of the sweep, by reducing each to a forced harmonic oscillator whose kernel is non-negative precisely because the rotation angle is $\pi/2$; consequently $V = |N|$ and Q is tight.

At Σ the three hypotheses of the concavity method hold exactly: $Q(\Sigma) = A_R^*$ structurally, with no integral evaluated (Theorem 52); $\delta Q(\Sigma) = 0$ identically in Romik's constants, so his ODE families *are* the critical-point equations of Q (Theorem 45); and $\delta^2 Q < 0$.

The second variation is the technical core. Setting $p(t) = \eta(t)$ and $q(t) = \eta(t+\pi/2)$ identifies it with a 2×2 Sturm–Liouville system on $[0, \pi/2]$ whose first eigenvalue *is* the best constant,

$$c^* = 0.7309566\ldots,$$

certified above $\frac{73}{100}$ in ball arithmetic (Theorem 73). On the overlap of the two sign sets the coupling is a total derivative, which is why every pointwise estimate loses and this one does not. The same system, with the cell boundaries as parameters, gives concavity on *every* ordered anchored cell (Theorem 76), and a uniform constant $\frac{3}{10}$ on the mirror class (Theorem 78).

These combine into two optimality statements. Theorem 71 gives $|T| \le A_R^*$ for every ambidextrous sofa with cap data in the convex domain $\mathcal{D}$ cut out by the gauge, (RC), a separation inequality and Σ's sign pattern, with the quantitative form

$$|T| \;\le\; A_R^* - \tfrac{73}{100}\,\|H - H_\Sigma\|^2_{L^2(0,\pi)}$$

of Theorem 79 and uniqueness (Corollary 80). Proposition 88 exhibits an explicit C^2-ball of radius $\frac{1}{20}$ inside $\mathcal{D}$. Theorem 83 is the non-perturbative version: on the mirror class, with no smallness of $H - H_\Sigma$ assumed, $|T| \le A_R^* - \frac{3}{10}\|H - H_\Sigma\|^2$.

What is not proved, and why

Optimality of Σ remains open, and the obstructions are recorded rather than smoothed over. V is the exact flux of the advancing sweep, so $V \ge |N|$ always by Reynolds' transport theorem (Proposition 28) and Q bounds nothing where the sweep fails to be injective; the domain cannot be widened by a concave majorant of the excess, since none exists (Proposition 81), though a semiconcave one does (Theorem 82). The separation hypothesis is independent of (RC). Gerver's cap violates (RC), but that is not what excludes it: it is not the cap of any *connected* ambidextrous sofa (Remark 90). The sign hypothesis of Theorem 83 is permanent, not provisional: the incircle of the unit square is a connected ambidextrous sofa satisfying every other condition and violating it (Remark 86).

Claims are labelled throughout: VERIFIED means machine-checked in Lean 4 with no `sorry`, PROVED a complete human proof, HEURISTIC computational evidence only. Numerical values that carry weight are computed in ball or exact rational arithmetic with stated precision; Section 18 separates what is ours from what is Baek's and Romik's.

2 Setting

Write L for the closed L-shaped hallway of unit width with its corner at the origin, $\mu_t = (\cos t, \sin t)$, $\nu_t = (-\sin t, \cos t)$, and for a rotation path $c : [0, \pi/2] \to \mathbb{R}^2$ let

$$C_t = \{p : \langle p - c(t), \mu_t\rangle \le 1\} \cap \{p : \langle p - c(t), \nu_t\rangle \le 1\}, \qquad Q_t = \{p : \langle p - c(t), \mu_t\rangle < 0\} \cap \{p : \langle p - c(t), \nu_t\rangle < 0\},$$

so that the hallway in position t is $H_t = C_t \setminus Q_t$. The one-sided sofa is $S = \bigcap_t H_t$, and writing $C = \bigcap_t C_t$ and $U = \bigcup_t Q_t$ we have $S = C \setminus U$ with C convex.

Let $\rho(x, y) = (x, 1 - y)$ be the reflection in the line $y = \frac{1}{2}$. Romik's ambidextrous sofa is

$$\Sigma = S \cap \rho(S) = C_2 \setminus \big(U \cup \rho U\big), \qquad C_2 := C \cap \rho C \text{ convex}, \tag{1}$$

with $|\Sigma| = A_R^* = 1.6449552184\ldots$, and c is Romik's piecewise path with breakpoints at β and $\pi/2-\beta$, where

$$\beta = \arctan\Big(\tfrac{1}{2}\big(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\big)\Big) = 0.2896538208\ldots \tag{2}$$

Baek's argument for the one-corner problem builds an overestimating region whose area is a quadratic functional on a convex domain of convex bodies, and proves it concave by exhibiting it as a linear functional minus Mamikon regions, whose areas are convex quadratics. Concavity reduces global optimality to a first-order condition. The construction is organised around $S = K \setminus \mathcal{N}(K)$: one cap, one niche.

By (1) the ambidextrous analogue has *two* niches, and

$$|U \cup \rho U| = |U| + |\rho U| - |U \cap \rho U|. \tag{3}$$

The last term appears in the upper bound with a $+$ sign. Since concavity is obtained by *subtracting* convex quadratics, an added overlap term is not covered by the mechanism. The purpose of this note is Theorem 8: the term is zero.

3 The niches are disjoint

Lemma 1 (Niche ceiling)**.** *For every $t \in [0,\pi/2]$ we have $Q_t \subseteq \{(x,y) : y \le c_y(t)\}$.*

Proof. Q_t is the open convex cone with apex $c(t)$ spanned by $-\mu_t$ and $-\nu_t$, so $q \in Q_t$ means $q - c(t) = -a\mu_t - b\nu_t$ with $a, b > 0$. For $t \in [0,\pi/2]$ both $\sin t \ge 0$ and $\cos t \ge 0$, hence

$$q_y - c_y(t) = -a \sin t - b \cos t \le 0. \qquad \square$$

Corollary 2. *Put $M := \max_{t\in[0,\pi/2]} c_y(t)$. Then $U \subseteq \{y \le M\}$ and $\rho U \subseteq \{y \ge 1 - M\}$. If $M < \frac{1}{2}$ then $U \cap \rho U = \emptyset$.*

Proof. The first inclusion is Lemma 1 applied to each t; the second is its image under ρ. If $M < \frac{1}{2}$ then $1 - M > \frac{1}{2} > M$, so the two half-planes are disjoint. $\square$

It remains to evaluate M for Romik's path. On the middle piece $[\beta, \pi/2-\beta]$ Romik's path is

$$c(t) = R(t)v(t) + \kappa, \qquad v(t) = \begin{pmatrix} f_1 \cos\frac{t}{2} + f_2 \sin\frac{t}{2} - 1 \\ -f_2\cos\frac{t}{2} + f_1 \sin\frac{t}{2} - 1 \end{pmatrix}, \tag{4}$$

with $R(t)$ the rotation by t, and the two facts we use are

$$\kappa_y = \tfrac{1}{2} \quad \text{exactly}, \qquad f_2 = (1-\sqrt{2})\, f_1. \tag{5}$$

Proposition 3. *For $t \in [\beta, \pi/2-\beta]$,*

$$c_y(t) - \tfrac{1}{2} = f_1 \sin\tfrac{3t}{2} - f_2\cos\tfrac{3t}{2} - (\sin t + \cos t) = f_1\sqrt{4-2\sqrt{2}}\,\sin\Big(\tfrac{3t}{2} + \tfrac{\pi}{8}\Big) - \sqrt{2}\,\sin\Big(t + \tfrac{\pi}{4}\Big). \tag{6}$$

In particular c_y is symmetric about $t = \pi/4$ on this interval, and

$$M = c_y(\pi/4) = \tfrac{1}{2} - \Big(\sqrt{2} - f_1\sqrt{4-2\sqrt{2}}\Big). \tag{7}$$

Proof. By (4) and $\kappa_y = \frac{1}{2}$, $c_y(t) - \frac{1}{2} = \sin t\, v_x + \cos t\, v_y$. Write $S = \sin\frac{t}{2}$, $C = \cos\frac{t}{2}$, so $\sin t = 2SC$ and $\cos t = C^2 - S^2$. Expanding and collecting the coefficients of f_1 and f_2 gives

$$\sin t\, v_x + \cos t\, v_y = f_1 S(3C^2 - S^2) + f_2 C(3S^2 - C^2) - (\sin t + \cos t),$$

and the triple-angle identities $S(3C^2 - S^2) = \sin\frac{3t}{2}$, $C(3S^2 - C^2) = -\cos\frac{3t}{2}$ give the first equality in (6). For the second, substitute $f_2 = (1 - \sqrt{2})f_1$ from (5): the bracket becomes $\sin\frac{3t}{2} + (\sqrt{2} - 1)\cos\frac{3t}{2}$, of amplitude $\sqrt{1 + (\sqrt{2} - 1)^2} = \sqrt{4 - 2\sqrt{2}}$ and phase $\arctan(\sqrt{2} - 1) = \pi/8$.

Both sinusoids in (6) are invariant under $t \mapsto \pi/2 - t$: indeed $\sin(\frac{3}{2}(\pi/2 - t) + \pi/8) = \sin(7\pi/8 - \frac{3t}{2}) = \sin(\pi/8 + \frac{3t}{2})$ and $\sin(3\pi/4 - t) = \sin(\pi/4 + t)$. Hence c_y is symmetric about $\pi/4$, where both sinusoids attain the value 1 simultaneously; (7) follows. $\square$

That $\pi/4$ maximises c_y over all of $[0, \pi/2]$, and not merely over the middle piece, follows from the corresponding closed form on the outer pieces.

Proposition 4. *For $t \in [0, \beta]$,*

$$c_y(t) = \tfrac{1}{2} + a_1 \sin 2t - \sin t - \tfrac{1}{2}\cos t, \tag{8}$$

and c_y is strictly increasing there, so $\max_{[0,\beta]} c_y = c_y(\beta)$. By the symmetry $c_y(t) = c_y(\pi/2 - t)$ the same bound holds on $[\pi/2 - \beta, \pi/2]$. Consequently $M = c_y(\pi/4)$ as claimed in (7).

Proof. On $[0, \beta)$ Romik's path is $c(t) = R(t)v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1 \cos t - 1,\ a_1 \sin t - \frac{1}{2})$ and $\kappa_y^{(1)} = \frac{1}{2}$, so $c_y(t) - \frac{1}{2} = \sin t\, v_{1x} + \cos t\, v_{1y} = 2a_1 \sin t \cos t - \sin t - \frac{1}{2}\cos t$, which is (8). Differentiating,

$$c_y'(t) = 2a_1 \cos 2t - \cos t + \tfrac{1}{2}\sin t \ \geq\ 2a_1 \cos 2\beta - 1 \ =\ 0.4650\ldots > 0 \qquad (t \in [0, \beta]),$$

using $\cos 2t \geq \cos 2\beta$ and $\cos t - \frac{1}{2}\sin t \leq 1$ for $t \geq 0$. Hence c_y is strictly increasing on $[0, \beta]$, with $c_y(\beta) = 0.2143795161\ldots$, which is below $c_y(\pi/4) = 0.3878381292\ldots$. $\square$

Remark 5. Evaluating (8) and (6) at $t = \beta$ gives the same value to $4.4 \cdot 10^{-31}$ at 30-digit precision. Since the two expressions were derived from different pieces of Romik's path, this is an independent check on both, and on (10).

4 A closed form for f_1

Romik's constant f_1 is recorded in the literature as the decimal 1.202938908156911389. The following determines it in closed form. Let

$$a_1 = \tfrac{1}{4}\sqrt{\,4 + \sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}}\,} = 0.8752873624\ldots \tag{9}$$

Proposition 6.

$$f_1 = \frac{\frac{4}{3}\, a_1 \cos\beta}{\cos\frac{\beta}{2} + (1 - \sqrt{2})\sin\frac{\beta}{2}}. \tag{10}$$

Proof. On $[0,\beta)$ Romik's path is $c(t) = R(t)v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1 \cos t - 1,\ a_1 \sin t - \frac{1}{2})$ and $\kappa^{(1)} = (1 - a_1, \frac{1}{2})$; on $[\beta, \pi/2 - \beta]$ it is (4) with $\kappa = (1 - \frac{4}{3}a_1, \frac{1}{2})$. Both κ's have second coordinate $\frac{1}{2}$ and their first coordinates differ by exactly $\frac{1}{3}a_1$. Continuity at $t = \beta$ therefore gives $R(\beta)\big(v_1(\beta) - v(\beta)\big) = (-\frac{1}{3}a_1, 0)$, so

$$v_1(\beta) - v(\beta) = R(-\beta)\big(-\tfrac{1}{3}a_1, 0\big) = \big(-\tfrac{1}{3}a_1 \cos\beta,\ \tfrac{1}{3}a_1 \sin\beta\big).$$

Reading the first coordinate and using $f_2 = (1-\sqrt{2})f_1$,

$$\big(a_1 \cos\beta - 1\big) - \Big(f_1\big(\cos\tfrac{\beta}{2} + (1-\sqrt{2})\sin\tfrac{\beta}{2}\big) - 1\Big) = -\tfrac{1}{3}a_1 \cos\beta,$$

which rearranges to (10). □

Remark 7. Evaluating (10) at 50 digits gives $f_1 = 1.2029389081569113890702 23\ldots$, so the tabulated decimal is its rounding. As an independent check, the *second* coordinate of the junction relation, which was not used in the derivation, is satisfied to $1.7 \cdot 10^{-51}$ at the same precision.

5 Disjointness of the niches for Σ

Theorem 8. *For Romik's ambidextrous sofa, $U \cap \rho U = \emptyset$, and consequently*

$$|\Sigma| = |C_2| - |U| - |\rho U| = |C_2| - 2|U|.$$

Proof. By Proposition 3, $M < \frac{1}{2}$ is equivalent to $f_1\sqrt{4 - 2\sqrt{2}} < \sqrt{2}$, i.e. to

$$f_1^2 < \frac{2+\sqrt{2}}{2}. \tag{11}$$

With f_1 given by (10) in terms of the algebraic numbers (2) and (9), both sides of (11) are explicit algebraic numbers, and

$$f_1^2 = 1.4470620167577420940\ldots, \qquad \frac{2+\sqrt{2}}{2} = 1.7071067811865475244\ldots,$$

so (11) holds with margin $0.2600447644\ldots$. Corollary 2 gives $U \cap \rho U = \emptyset$, and then (3) together with (1). Finally $|\rho U| = |U|$ because ρ is an isometry. □

Corollary 9. *$M = 0.3878381292441942964\ldots$, the two niches are separated by the horizontal band of width $1 - 2M = 0.2243237415116114\ldots$ about $y = \frac{1}{2}$, and the ambidextrous upper bound requires only* one *niche to be modelled: one core and two tails, the same shape as in the one-corner problem.*

6 The quotient reduction and a conditional statement

Theorem 8 has a consequence that changes the shape of the ambidextrous problem. Since $U \subseteq \{y \le M\}$ with $M < \frac{1}{2}$, the whole lower niche lies below the symmetry axis, so intersecting with the lower half-strip deletes ρU outright.

Corollary 10 (Quotient reduction)**.** *Put* $K^- := C_2 \cap \{y \le \frac{1}{2}\}$*, a convex body. Then*

$$\Sigma \cap \{y \le \tfrac{1}{2}\} = K^- \setminus U, \qquad |\Sigma| = 2\big(|K^-| - |U \cap K^-|\big).$$

That is a convex cap minus *one* niche: the shape Baek's argument is built on. Numerically, at $n = 1441$ hallways, $|\Sigma| = 1.645059395$ and $|\Sigma \cap \{y \le \frac{1}{2}\}| = 0.822529698$, whose double reproduces $|\Sigma|$ to $3.8 \cdot 10^{-15}$.

Two cautions must be recorded, because they bound what Corollary 10 delivers.

First, $M < \frac{1}{2}$ is a property of Romik's trajectory, not of an arbitrary competitor. A competitor with $M \ge \frac{1}{2}$ has overlapping niches, and by (3) overlap *increases* $|\Sigma|$; so it cannot be assumed away on extremal grounds.

Second, and this is what makes the reduction usable, connectedness does appear to exclude it, but not for the reason one first tries. A *single* pair $Q_t \cup \rho Q_t$ removes a full vertical segment, and hence disconnects any body meeting both sides of it, exactly when the apex height exceeds a threshold $h^*(t)$; and computation gives $h^*(t) > \frac{1}{2}$ throughout $(0, \pi/2)$, with $h^*(t) \to \frac{1}{2}$ as $t \to 0$, $h^*(\pi/4) = 0.5100$ and $h^*(1.52) = 0.6967$. So no single-t argument forces $M \le \frac{1}{2}$; all it gives is the pointwise constraint $c_y(t) \le h^*(t)$.

In fact a single pair does force it, and the earlier reading to the contrary was an artifact of testing the vertical line at a fixed distance from the apex.

Theorem 11 (Connectedness ceiling)**.** *Let* $p = c(t_0)$ *with* $t_0 \in (0, \pi/2)$ *and suppose* $p_y > \frac{1}{2}$*. Put*

$$\varepsilon_0 := \frac{2p_y - 1}{2 \tan t_0} > 0.$$

Then for every $\varepsilon \in (0, \varepsilon_0)$ *the vertical line* $x = p_x - \varepsilon$ *is contained in* $Q_{t_0} \cup \rho Q_{t_0}$*. Hence any ambidextrous moving sofa omits the open vertical strip* $\{p_x - \varepsilon_0 < x < p_x\}$*, and a connected one lies entirely in one of the two half-planes it separates.*

Proof. Fix $\varepsilon > 0$ and consider $q = (p_x - \varepsilon, y)$. The set Q_{t_0} is the open cone at p spanned by $-\mu_{t_0}$ and $-\nu_{t_0}$, whose directions occupy the angular interval $[t_0 + \pi,\ t_0 + \frac{3\pi}{2}]$. Writing $h := p_y - y$, the direction $q - p = (-\varepsilon, -h)$ has angle $\pi + \arctan(h/\varepsilon)$ when $h > 0$, so $q \in Q_{t_0}$ iff $\arctan(h/\varepsilon) \ge t_0$, that is iff

$$y \le p_y - \varepsilon \tan t_0.$$

Applying ρ and using $\rho q - \rho p = R(q - p)$ with $R = \operatorname{diag}(1, -1)$, the reflected cone ρQ_{t_0} has apex $(p_x, 1 - p_y)$ and occupies the angular interval $[\frac{\pi}{2} - t_0,\ \pi - t_0]$, giving by the same computation

$$q \in \rho Q_{t_0} \iff y \ge 1 - p_y + \varepsilon \tan t_0.$$

The two sets therefore cover the whole line exactly when $p_y - \varepsilon \tan t_0 \ge 1 - p_y + \varepsilon \tan t_0$, i.e. when $2\varepsilon \tan t_0 \le 2p_y - 1$, i.e. when $\varepsilon \le \varepsilon_0$. Since $p_y > \frac{1}{2}$ we have $\varepsilon_0 > 0$ and the range is nonempty. Taking the union over $\varepsilon \in (0, \varepsilon_0)$ gives the strip. ☐

Remark 12. The criterion is sharp in both directions: if $p_y \le \frac{1}{2}$ then $\varepsilon_0 \le 0$ and the two cones leave a gap at *every* distance from the apex, which is why $\frac{1}{2}$ is exactly the threshold. It also explains a numerical trap: ε_0 can be very small (at $t_0 = 1.3$, $p_y = 0.55$ it is 0.0139), so sampling the line at a fixed distance such as 0.02 detects no covering and suggests, wrongly, that a single pair never separates. The criterion was checked against direct computation at seven pairs (t_0, p_y) on both sides of ε_0, and at three pairs with $p_y < \frac{1}{2}$.

Granting that an ambidextrous sofa cannot lie in one of the two half-planes separated by such a strip, since it must meet both arms of the hallway H_{t_0} whose inner corner is p, Theorem 11 yields $\max_t c_y(t) \leq \frac{1}{2}$ for every connected ambidextrous moving sofa, and the hypothesis of the next proposition is automatic.

Proposition 13 (Conditional form)**.** *Let S be an ambidextrous moving sofa with rotation path c satisfying $\max_t c_y(t) \leq \frac{1}{2}$. Then its two niches are disjoint and $|S| = 2(|K^-| - |U \cap K^-|)$ with K^- convex and U a single niche. In particular the area functional of the ambidextrous problem restricted to this class is of the cap-minus-one-niche form to which the concavity method applies. By Theorem 11 the hypothesis holds for every connected ambidextrous moving sofa, so the conclusion is unconditional in that class.*

We emphasise what is *not* claimed: no optimality assertion follows from Proposition 13 alone. Baek's route additionally requires an existence and regularity step (a maximum monotone sofa of rotation angle $\pi/2$ satisfying an injectivity condition), and his balancing argument for that step does not transfer, for a reason worth recording. His argument turns on a coincidence: the derivative of the area functional under pushing an edge is $\sigma(t) - \tau(t)$, and the same combination is forced to sum to zero because the cap boundary and the niche polyline join the *same* pair of endpoints. In the quotient, ∂K^- acquires the horizontal cut at $y = \frac{1}{2}$, where the polyline contributes nothing, and the identity becomes $\sum_{t \neq \text{cut}} (\sigma - \tau)(v_t \cdot u_0) = \sigma_{\text{cut}} > 0$, which is consistent with $\sigma \leq \tau$ throughout. One obtains an inequality where Baek obtains an equality, and it is the equality that his argument consumes.

7 A relation between Romik's constants

The phase junctions of Σ admit a characterisation that ties two of Romik's constants together. Recall a_1 from (9) and β from (2).

Theorem 14. *$4a_1 \sin\beta = 1$.*

Proof. Write $u := \sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}$, so that $\tan\beta = u/2$ by (2) and hence $\sin\beta = u/\sqrt{4+u^2}$, and $w := \sqrt[3]{71+8\sqrt{2}} + \sqrt[3]{71-8\sqrt{2}}$, so that $a_1 = \frac{1}{4}\sqrt{4+w}$ by (9). Then

$$4a_1 \sin\beta = \frac{u\sqrt{4+w}}{\sqrt{4+u^2}},$$

so the assertion is equivalent to $u^2(4+w) = 4 + u^2$, that is to

$$u^2\,(3+w) = 4. \tag{12}$$

Put $p := \sqrt[3]{\sqrt{2}+1}$ and $q := \sqrt[3]{\sqrt{2}-1}$. Then $pq = 1$ and $p^3 - q^3 = 2$, so

$$u^3 = (p-q)^3 = p^3 - q^3 - 3pq(p-q) = 2 - 3u,$$

i.e.

$$u^3 + 3u = 2. \tag{13}$$

Set $x := u^2$. Then (13) reads $u(x+3) = 2$; squaring and using $u^2 = x$ gives

$$x(x+3)^2 = 4, \qquad \text{i.e.} \qquad x^3 + 6x^2 + 9x - 4 = 0. \tag{14}$$

Now let $W := 4/x - 3$. Clearing denominators,

$$W^3 - 51W - 142 = \frac{(4-3x)^3 - 51(4-3x)x^2 - 142x^3}{x^3} = \frac{-16\big(x^3 + 6x^2 + 9x - 4\big)}{x^3} = 0$$

by (14), the middle equality being the polynomial identity $(4-3x)^3 - 51(4-3x)x^2 - 142x^3 = -16(x^3 + 6x^2 + 9x - 4)$.

On the other hand w satisfies the same cubic: with $A := \sqrt[3]{71 + 8\sqrt{2}}$, $B := \sqrt[3]{71 - 8\sqrt{2}}$ one has $AB = \sqrt[3]{71^2 - 128} = \sqrt[3]{4913} = 17$ and $A^3 + B^3 = 142$, so $w^3 = A^3 + B^3 + 3AB\,w = 142 + 51w$. The cubic $W^3 - 51W - 142$ has discriminant $-4(-51)^3 - 27(-142)^2 = 530604 - 544428 = -13824 < 0$ and therefore exactly one real root; both $4/x - 3$ and w are real roots, so $4/x - 3 = w$. This is (12). $\square$

Corollary 15. *$2\tan\beta$ is the real root of $u^3 + 3u = 2$, equivalently $4\tan^3\beta + 3\tan\beta = 1$, the cubic already recorded in the literature, and*

$$\sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}} = u^4 + 6u^2 + 6.$$

Proof. The first assertion is (13). For the second, (14) gives $(x+3)^2 = 4/x$, so $w = 4/x - 3 = (x+3)^2 - 3 = x^2 + 6x + 6$ with $x = u^2$. $\square$

Proof of the junction statement. On $[0, \beta)$ Romik's path is $c(t) = R_t v(t) + \kappa^{(1)}$ with $v(t) = (a_1 \cos t - 1,\ a_1 \sin t - \frac{1}{2})$. Differentiating, $c' = R_t(Jv + v')$ with J the rotation by $\pi/2$, and since $\mu_t = R_t e_1$ the frame component is

$$c'(t) \cdot \mu_t = (Jv + v')_x = -v_y - a_1 \sin t = -\big(a_1 \sin t - \tfrac{1}{2}\big) - a_1 \sin t = \tfrac{1}{2} - 2a_1 \sin t.$$

This vanishes precisely when $\sin t = 1/(4a_1)$. On the other hand the phase boundary of Σ at $t = \beta$ is where the SOL1 phase ends, and by Proposition 4 the two closed forms for c_y agree there; the same computation on the SOL5 phase gives $c' \cdot \nu_t = 2a_1 \cos t - \frac{1}{2}$, vanishing when $\cos t = 1/(4a_1)$, i.e. at $t = \pi/2 - \beta$. Since the two junctions are β and $\pi/2 - \beta$ and $\cos(\pi/2 - \beta) = \sin\beta$, both conditions reduce to $\sin\beta = 1/(4a_1)$. $\square$

Remark 16 (Attribution)**.** Neither (13) nor (14) is new. The standard description of Σ gives its area as $x + \arctan y$ where $x^2(x+3) = 8$ and $y(4y^2 + 3) = 1$; the second of these is exactly (13) with $y = \tan\beta = u/2$, and the first is (14) shifted, since $x = u^2 + 1$ turns $x^3 + 3x^2 - 8$ into $u^6 + 6u^4 + 9u^2 - 4$. Equivalently

$$A_R^* = u^2 + 1 + \arctan(u/2), \qquad u^3 + 3u = 2,$$

so a single algebraic number generates both parts of the area. What we have not found stated is the relation $4a_1 \sin\beta = 1$ of Theorem 14 between the shape constant a_1 and β, nor $w = u^4 + 6u^2 + 6$; but both are elementary consequences of the two standard cubics, and we make no priority claim. Every step was checked numerically at 60 digits, and the two polynomial steps are machine-checked (`u_sq_cubic`, `cardano_substitution`).

Theorem 14 has a consequence for the transfer of Baek's method. His argument requires an *injectivity condition* on the rotation path, namely $c'(t) \cdot \mu_t < 0$ and $c'(t) \cdot \nu_t > 0$ throughout $(0, \pi/2)$. By the computation above, for Σ these quantities vanish exactly at β and $\pi/2 - \beta$ and have the wrong sign outside $[\beta, \pi/2 - \beta]$: at $t = 0.05$ one finds $c' \cdot \mu_t = +0.4125$. So Σ satisfies the injectivity condition on its middle phase and violates it on both outer phases, and the failure boundary is exactly the phase junction.

Remark 17 (Where the difficulty of the ambidextrous problem sits)**.** The outer phases $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$ are also where the contact arcs of $\partial\Sigma$ are *constant*, pinned at the ρ-fixed point $(1, \frac{1}{2})$ to $3 \cdot 10^{-31}$: there the supporting hallway touches Σ along a segment rather than at a point. Three separate obstructions therefore coincide on those phases: the contact structure degenerates, second variation arguments lose the contact point, and the injectivity hypothesis of the one-corner proof fails.

8 The ambidextrous problem as a single family of angle range π

Finally we record a reformulation which explains why the transfer is delicate. Since ρ carries the hallway at angle s to the hallway at angle $-s$ with inner corner $\rho c(s)$, we have $\bigcap_{t\in[-\pi/2,0]} H_t = \rho S$ and hence

$$\Sigma = \bigcap_{t\in[-\pi/2,\pi/2]} H_t, \qquad c(-t) = \rho c(t). \tag{15}$$

Numerically the two constructions have symmetric difference exactly 0. So the ambidextrous constraint family is a *single* family spanning an angle range of π, rather than two families of range $\pi/2$; and its corner path is discontinuous at $t = 0$, where $c_y(0) = 0$ while $\rho c(0)$ has height 1. Both features lie outside the setting of the one-corner argument, which is developed for rotation angle $\omega \le \pi/2$ and whose maximisers attain $\omega = \pi/2$. Equation (15) is thus a reformulation, not a reduction: it exhibits the ambidextrous problem inside a larger class rather than inside the solved one.

9 The niche area in convex-linear data

The obstruction recorded in Remark 17 is that the one-corner argument's injectivity condition fails for Σ on the two outer phases. We now show that the failure is an artefact of how the condition is stated, and derive an exact formula for the niche area whose every ingredient is convex-linear in the cap.

Throughout, K is the cap, $h = h_K$ its support function, and

$$F(t) := h(\mu_t), \qquad G(t) := h(\nu_t).$$

Lemma 18 (The corner is affine in the support function)**.** *Place the hallway at angle t and push both outer walls inward until they touch K. Then the inner corner sits at*

$$c(t) = \big(F(t) - 1\big)\mu_t + \big(G(t) - 1\big)\nu_t, \tag{16}$$

and the removed wedge is $Q_t = \{\langle p, \mu_t\rangle < F(t) - 1\} \cap \{\langle p, \nu_t\rangle < G(t) - 1\}$.

Proof. The two outer walls are the lines with outer normals μ_t, ν_t; touching K means their offsets are $F(t), G(t)$. The corner lies one unit inside each, so $\langle c, \mu_t\rangle = F - 1$ and $\langle c, \nu_t\rangle = G - 1$, which is (16) since (μ_t, ν_t) is orthonormal. The wedge is the intersection of the two inner half-planes through c. $\square$

Since $h \mapsto h$ is linear under Minkowski sum, (16) makes $c(t)$, and hence each of

$$\alpha_1(t) := -\langle c'(t), \mu_t\rangle = G(t) - 1 - F'(t), \tag{17}$$
$$\alpha_2(t) := \quad \langle c'(t), \nu_t\rangle = F(t) - 1 + G'(t), \tag{18}$$
$$\sigma(t) := c_y(t)/\cos t = \big(F(t) - 1\big)\tan t + G(t) - 1, \tag{19}$$

an affine-linear functional of h, hence convex-linear on a domain of convex bodies closed under Minkowski sum. The wedge Q_t has two faces, the rays $c - s\nu_t$ (face 1) and $c - s\mu_t$ (face 2), and α_1, α_2 are the signed distances from c to the envelope points on them: the one-corner injectivity condition is exactly $\alpha_1, \alpha_2 > 0$.

Lemma 19 (Normal velocities)**.** *The face-*1 *line advances in its own normal direction with speed* $s - \alpha_1(t)$ *at the point at distance* s *from the corner, and the face-*2 *line with speed* $\alpha_2(t) - s$*.*

Proof. Face 1 lies on $\ell_t = \{\langle p, \mu_t\rangle = F(t) - 1\}$. For p fixed the signed gap to ℓ_t is $F(t) - 1 - \langle p, \mu_t\rangle$, whose t-derivative is $F'(t) - \langle p, \nu_t\rangle$ because $\mu_t' = \nu_t$. At $p = c - s\nu_t$ we have $\langle p, \nu_t\rangle = G - 1 - s$, so the speed is $F' - G + 1 + s = s - \alpha_1$ by (17). Face 2 lies on $\{\langle p, \nu_t\rangle = G(t) - 1\}$ and $\nu_t' = -\mu_t$, so the same computation gives $G' + \langle p, \mu_t\rangle = G' + F - 1 - s = \alpha_2 - s$ at $p = c - s\mu_t$. □

So the niche is created by the *inner* part of face 2 and the *outer* part of face 1, and *no sign hypothesis on* α_1 *is involved*: where $\alpha_1 < 0$ the whole visible face 1 advances, and where $\alpha_2 \le 0$ face 2 contributes nothing. This is the promised repair of Remark 17: each face is used only on the range where its own arm has the right sign, and the reported failure came from requiring both arms simultaneously.

Write W_2 for the region swept by the segments $[c(t), c(t) - \alpha_2(t)^+\mu_t]$, and W_1^{out} for the region swept by the face-1 segments running from $c(t) - \alpha_1(t)^+\nu_t$ outward to the corridor floor $\{y = 0\}$; note the face-1 direction $-\nu_t = (\sin t, -\cos t)$ points downward, and the floor is reached at distance exactly $\sigma(t)$, which is why (19) is the relevant reach.

Proposition 20 (Niche area in convex-linear data)**.** *If the two sweeps are disjoint, injective and cover the niche, then*

$$|N| = \int_0^{\pi/2} \Big[\tfrac{1}{2}\big(\alpha_2^+\big)^2 + \tfrac{1}{2}\big(\sigma - \alpha_1\big)^2 - \tfrac{1}{2}\big(\alpha_1^-\big)^2\Big]\,dt. \tag{20}$$

Proof. For the face-2 sweep $\Phi(s,t) = c(t) - s\mu_t$ one has $\partial_s\Phi = -\mu_t$ and $\partial_t\Phi = c' - s\nu_t$, so $\det[\partial_s\Phi, \partial_t\Phi] = -\langle c', \nu_t\rangle + s = s - \alpha_2$, and integrating $|s - \alpha_2|$ over $s \in [0, \alpha_2^+]$ gives $\frac{1}{2}(\alpha_2^+)^2$. For the face-1 sweep $\Psi(s,t) = c(t) - s\nu_t$ one gets $\det = s - \alpha_1$, and

$$\int_{\alpha_1^+}^{\sigma} (s - \alpha_1)\,ds = \tfrac{1}{2}(\sigma - \alpha_1)^2 - \tfrac{1}{2}(\alpha_1^-)^2,$$

which is $\frac{1}{2}(\sigma - \alpha_1)^2$ when $\alpha_1 \ge 0$ and $\frac{1}{2}\sigma^2 - \alpha_1\sigma$ when $\alpha_1 < 0$, as required. Injectivity turns each $\int |\det|$ into the area of the image, disjointness makes the two areas add, and covering makes the sum $|N|$. □

Remark 21 (What is measured, and what it gives)**.** For Σ the three hypotheses of Proposition 20 are verified numerically and sharply: the two sweeps have intersection area 0 and leave 0 of the niche uncovered, to the printing precision of the polygon oracle. The right-hand side of (20), evaluated with exact derivatives of SOL1/SOL5/SOL6 and Gauss–Legendre quadrature on each phase separately, is

$$V = 0.184193197089\ldots,$$

stable in the twelfth digit from 20 to 320 nodes per phase, against a polygonal measurement 0.184193171 of the same region, and an inscribed quadrilateral union under-measures, so the two agree to $2.6\cdot 10^{-8}$ from the expected side. The implied cap area $A_R^* + 2V = 2.0133416\ldots$ lies just below the circumscribed polygonal values 2.0133455 and 2.0133420 at 241 and 481 constraints, which must decrease to it. We therefore record (20) as an identity for Σ at the 10^{-8} level.

Two of the three terms of (20) are convex quadratics in h, since $x \mapsto \frac{1}{2}(x^+)^2$ and $x \mapsto \frac{1}{2}x^2$ are convex and $\alpha_2, \sigma - \alpha_1$ are affine in h. This is what the generalised Mamikon theorem buys, and it is the reason (20) is the right shape: for $|\Sigma| = |C_2| - 2|N|$ a convex $|N|$ gives a concave upper bound, for which a first-order condition suffices.

Proposition 22 (The obstruction, in closed form)**.** *The third term of* (20) *is subtracted, so it contributes a* convex *term to* $|C_2| - 2|N|$ *and is the one ingredient with the wrong curvature. It is supported exactly on the degenerate phase* $[0, \beta)$*, where* $\alpha_1 < 0$*, and there* $\alpha_1(t) = 2a_1 \sin t - \frac{1}{2}$*, so*

$$\frac{1}{2}\int_0^{\pi/2} \left(\alpha_1^-\right)^2 dt = \frac{\beta}{8} - a_1(1 - \cos\beta) + a_1^2\Big(\beta - \tfrac{1}{2}\sin 2\beta\Big) = 0.011950270059\ldots, \tag{21}$$

which is $6.488\ldots\%$ *of* V*.*

Proof. On $[0, \beta)$ the path is SOL1 with $a_2 = 0$, so in complex form $z(t) = a_1 e^{2it} - (1 + \frac{i}{2})e^{it} + (1 - a_1 + \frac{i}{2})$ and $e^{-it}z'(t) = 2ia_1 e^{it} - i(1 + \frac{i}{2})$, whence $\alpha_1 = -\operatorname{Re}(e^{-it}z') = 2a_1 \sin t - \frac{1}{2}$. This is negative exactly for $\sin t < 1/(4a_1)$, i.e. for $t < \beta$ by Theorem 14. Expanding $\frac{1}{2}\int_0^\beta (\frac{1}{2} - 2a_1 \sin t)^2 dt$ and using $\int_0^\beta \sin^2 = \frac{\beta}{2} - \frac{1}{4}\sin 2\beta$ gives (21). □

Remark 23 (Where this leaves the problem)**.** Two obstructions have been removed and one remains. The first was that no trial underestimate of the niche was tight: the best previous one, built from the apex path, left a slack of 0.087 against $|\Sigma| = 1.645$. Formula (20) is tight, so that slack is gone. The second was that the injectivity condition fails for Σ on $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$; Lemma 19 shows the condition is not needed, only the two one-sided statements α_2^+ and $\sigma \geq \alpha_1^+$, and the second holds throughout with equality only at $t = \pi/2$.

What remains is (21). It is a subtracted square of a convex-linear quantity, i.e. a $\mathcal{J}$-type term of the same kind the one-corner argument must itself control, and it is localised on the phase where Σ degenerates. Establishing or refuting concavity of $|C_2| - 2V$ is a Hessian computation on the space of support-function perturbations, and is not attempted here. Neither is the second half of the argument, which would have to show that the three hypotheses of Proposition 20 hold for competitors and not only for Σ. So *optimality of* Σ *remains open*; what is new is that the upper bound is now tight and expressed entirely in convex-linear data, with a single explicitly computed term of the wrong sign.

10 The upper bound at Σ: tight, critical, concave

Since $\nu_t = \mu_{t+\pi/2}$, everything above is a functional of the single function

$$H(\theta) := h_K(\mu_\theta), \qquad \theta \in [0, \pi], \qquad F(t) = H(t), \quad G(t) = H(t + \pi/2).$$

Proposition 24 (The cap is an exact quadratic form)**.** *Let* $C = \bigcap_{t\in[0,\pi/2]} C_t$ *and* $C_2 = C \cap \rho C$. *Then*

$$|C_2| = \int_0^\pi \big(H(\theta)^2 - H'(\theta)^2\big)d\theta - H(0) - H(\pi). \tag{22}$$

Proof. ρ is a reflection in the line $y = \frac{1}{2}$, not in the origin: $\rho p = Rp + (0,1)$ with $R = \mathrm{diag}(1,-1)$, so $h_{\rho A}(u) = h_A(Ru) + u_y$. Since C is unbounded downward, $h_C = +\infty$ on the lower half-circle and $h_{\rho C} = +\infty$ on the upper one, so the support function of C_2 is

$$h_2(\theta) = H(\theta) \;\; (\theta \in [0,\pi]), \qquad h_2(\theta) = H(-\theta) + \sin\theta \;\; (\theta \in [-\pi, 0]).$$

For any support function in H^1 of the circle, $|C_2| = \frac{1}{2}\int h_2\, d\sigma_2$ with $\sigma_2 = (h_2 + h_2'')d\theta$ as a measure, and the distributional identity $\langle h_2'', h_2\rangle = -\int h_2'^2$ holds with no boundary or atom terms because the circle has no boundary. Splitting at $\theta = 0, \pi$ and substituting the two branches,

$$|C_2| = \int_0^\pi \Big[H^2 - H'^2 - H\sin s + H'\cos s - \tfrac{1}{2}\cos 2s\Big]ds,$$

and $\int_0^\pi \cos 2s\, ds = 0$ while $\int_0^\pi H'\cos s\, ds = -H(0) - H(\pi) + \int_0^\pi H \sin s\, ds$, which gives (22). $\square$

Note $-H(0) - H(\pi)$ is *linear*, so it does not enter the Hessian; and (22) is invariant under $H \mapsto H + v\cos\theta$, as it must be, that being a horizontal translation. Numerically (22) gives 2.013341613 against $A_R^* + 2V = 2.013341613$, agreeing to $9.6\cdot 10^{-13}$, an independent analytic confirmation of the whole chain $|\Sigma| = |C_2| - 2|N|$ with $|N| = V$, since (22) and (20) share no computation. Equivalently, writing $Q := |C_2| - 2V$,

$$Q(\Sigma) = 1.644955218425\ldots = A_R^* \quad \text{to } 5\cdot 10^{-13}. \tag{23}$$

Proposition 25 (The second variation)**.** *Fix the gauge* $H(0) = 1$ *(horizontal translation) and* $H(\pi/2) = 1$ *(unit corridor; this is also exactly what makes* σ *integrable against* $\tan t$ *at* $t = \pi/2$*), so that admissible* η *satisfy* $\eta(0) = \eta(\pi/2) = 0$. *Then, with* $E_1 = \{\alpha_1 < 0\}$ *and* $E_2 = \{\alpha_2 > 0\}$,

$$\begin{aligned}\tfrac{1}{2}\,\delta^2 Q[\eta] = \int_0^\pi (\eta^2 - \eta'^2)\, d\theta - \int_{E_2}\big(\eta(t) + \eta'(t + \tfrac{\pi}{2})\big)^2 dt \\ - \int_0^{\pi/2}\big(\eta(t)\tan t + \eta'(t)\big)^2 dt + \int_{E_1}\big(\eta(t+\tfrac{\pi}{2}) - \eta'(t)\big)^2 dt.\end{aligned} \tag{24}$$

Its principal part is negative definite: the coefficient of $\eta'(\theta)^2$ *is* -1 *on* $[0,\beta)$ *and on* $[\pi - \beta, \pi]$, *and* -2 *in between. Hence* (24) *is bounded above and has at most finitely many non-negative eigenvalues.*

Proof. $\delta\alpha_1 = \eta(t + \frac{\pi}{2}) - \eta'(t)$, $\delta\alpha_2 = \eta(t) + \eta'(t + \frac{\pi}{2})$ and $\delta\sigma = \eta(t)\tan t + \eta(t + \frac{\pi}{2})$ by (17)–(19); in the middle term the $\eta(t + \pi/2)$ cancels, leaving $\delta(\sigma - \alpha_1) = \eta(t)\tan t + \eta'(t)$. The integrand of (20) is C^1 across $\alpha_1 = 0$ and $\alpha_2 = 0$, so the moving boundaries of E_1, E_2 contribute nothing at second order, and δ^2 of each square is twice the square of the first variation. The cap contributes $2\int(\eta^2 - \eta'^2)$ by Proposition 24. For the principal part, collect: -1 from the cap, -1 from the σ term on $[0, \pi/2]$, $+1$ from the E_1 term on $[0,\beta)$, and -1 from the E_2 term at $\theta = t + \pi/2 \in [\pi/2, \pi - \beta)$. $\square$

Remark 26 (Measurements, and exactly what they do and do not give)**.** In a hat basis with $\eta(0) = \eta(\pi/2) = 0$:

Criticality. $\delta Q[\eta] = 0$ in every direction, $\|\delta Q\|_\infty/h = 1.8 \cdot 10^{-8}$ by central differences on Q itself. So Romik's ODEs are the Euler–Lagrange equations of Q. (An analytic assembly of the gradient first reported a nonzero value on $[0, \beta)$ and $[\pi/2, \pi/2 + \beta)$; that was a sign error on $\delta\big(-\frac{1}{2}(\alpha_1^-)^2\big) = +\alpha_1^-\delta\alpha_1$, caught by the finite-difference check.)

Concavity, on Σ's cell only. With Σ's own sign pattern $E_1 = [0, \beta)$, $E_2 = [0, \pi/2 - \beta)$, the form (24) is negative definite: $\sup \frac{1}{2}\delta^2 Q[\eta]/\|\eta\|^2_{L^2(0,\pi)} = -0.7358, -0.7339, -0.7331$ at $m = 32, 64, 128$, against the exact mass matrix. Under Baek's injectivity condition, which is exactly $E_1 = \emptyset$, $E_2 = [0, \pi/2]$, it is -0.8751. But (24) is *not* negative for every sign pattern: with $E_1 = E_2 = [0.4, 1.2]$ it is $+0.4123$, and with $E_1 = E_2$ a union of three intervals $+0.4225$. An earlier scan over intervals *anchored at* 0 suggested that $\tau_1 \le \tau_2$ suffices; the unanchored cases refute that and the suggestion is **retracted**.

Since α_1, α_2 are affine in H, each pointwise sign condition is a half-space and Σ's cell is convex. Concavity plus criticality on a convex set gives that Σ maximises Q *on that cell.* Nothing global follows.

What is missing, and it is worse than missing. All of the above concerns Q alone. The inequality $Q \ge |\text{sofa}|$, which is the only reason Q is relevant, **fails**; see Proposition 28. All numbers in this remark are floating point.

Remark 27 (The boundary of the cap, and where the sweep terminates)**.** The density $H + H''$ of the surface measure determines ∂C_2 completely: it vanishes on $(-\beta, \beta)$, where ∂C_2 is the single vertex $P = (1, \frac{1}{2})$; equals $\frac{1}{2}$ on $(\pi/2 - \beta, \pi/2 + \beta)$, where ∂C_2 is a circular arc of radius exactly $\frac{1}{2}$ centred at $(1 - 2a_1, \frac{1}{2})$; has an atom of mass 1.167050 at $\theta = \pi/2$, the corridor ceiling $y = 1$, a facet; vanishes again on $(\pi - \beta, \pi]$, the second vertex; and is a smooth arc in between. The ρ-image of the ceiling facet is the corridor floor, the segment $[-0.750575, 0.416475] \times \{0\}$, whose left endpoint lies directly below the centre of the radius-$\frac{1}{2}$ arc.

This settles the geometry of the outer arm of Proposition 20. Because C_2 is *convex*, the face-1 segment lies in C_2 as soon as its far endpoint does, so the claim $S_1 = \sigma$ reduces to a one-parameter containment: the endpoint $x(t) = c_x(t) + \sigma(t) \sin t$ must lie in the floor facet. It runs monotonically from $x(0) = 0$ to $x(\pi/2) = 0.416475$, the right end of the facet, and

$$x(\pi/2) = c_x(\pi/2) + \alpha_1(\pi/2) = \big(c_x(0) - \alpha_2(0)\big) + \big(\alpha_2(0) + \alpha_1(\pi/2) - (G(\pi/2) - 1)\big)$$

is an algebraic identity, the right-hand side being the left end of the facet plus its length $H'(\pi/2^+) - H'(\pi/2^-)$. So the tangency at $t = \pi/2$ is exact, not coincidental, and what remains of $S_1 = \sigma$ is the monotonicity of the explicit function x.

Proposition 28 (V over-estimates the niche, so Q is not an upper bound)**.** *For every admissible H, $V(H) \ge |N(H)|$. Consequently $Q = |C_2| - 2V \le |C_2| - 2|N|$, and for the maximal sofa $T_{\max} = C_2 \setminus (U \cup \rho U)$ with that cap, $|T_{\max}| = |C_2| - 2|N| \ge Q$. So Q is an upper bound for $|T_{\max}|$ only where $V = |N|$.*

Proof. V is precisely the flux of the advancing part of ∂Q_t, by Lemma 19 and the Jacobian computation in Proposition 20. By Reynolds' transport theorem the area gained by the sweeping region $N_T = \bigcup_{t \le T} Q_t \cap C_2$ satisfies $\frac{d}{dT}|N_T| \le \int_{\text{advancing part of } \partial Q_T} v_n \, ds$, because advance into already-swept territory contributes to the flux but gains no area. Integrating over $T \in [0, \pi/2]$ gives $|N| \le V$. □

Remark 29 (The size of the gap, measured)**.** Equality in Proposition 28 holds exactly when nothing is re-swept, which is what the three hypotheses of Proposition 20 assert, and which is why $V(\Sigma) = |N(\Sigma)|$. Perturbing H by a smooth bump η supported where $H + H'' \geq \frac{1}{2}$ (so convexity survives) and measuring $|N|$ with a polygon oracle against V from the same H, the excess $V - |N|$, corrected for the oracle's known $O(1/n)$ bias, is

ε	-0.20	-0.10	-0.05	0.05	0.10	0.20
$V - \|N\|$	$5.5 \cdot 10^{-3}$	$1.4 \cdot 10^{-4}$	$4.2 \cdot 10^{-6}$	$9.3 \cdot 10^{-5}$	$8.4 \cdot 10^{-4}$	$5.0 \cdot 10^{-3}$

strictly positive on *both* sides and growing faster than ε^2. So Σ is an isolated zero of $V - |N|$: the functional Q touches the true bound at Σ and falls strictly below it in every direction. Therefore Q cannot certify Σ, and the fact that Q is tight, critical and concave there (Remark 26) does not yield an optimality statement.

This locates the difficulty exactly. The repair is the one the one-corner argument performs: replace the exact flux V by the flux over a sub-family on which the sweep is *provably* injective, so that the integral is the area of a genuine subset of N and the inequality runs the right way, while the restriction is vacuous at Σ. Producing such a sub-family is the content of that argument's Chapters 3–6 and is not done here.

One reformulation that may help. A point p lies on the face-2 line at parameter t iff $\phi(t) := \langle p, \nu_t \rangle - (G(t) - 1)$ vanishes, and with $s(p,t) := F(t) - 1 - \langle p, \mu_t \rangle$ one computes $\phi' = s - \alpha_2$ and $s' = -\phi - \alpha_1$, i.e. in complex form

$$z := s + i\phi, \qquad z' = iz - (\alpha_1 + i\alpha_2).$$

So the whole face-2 sweep is governed by one linear ODE whose homogeneous part is an exact rotation, $z(t) = \zeta(t) + e^{it} w_0$ with w_0 an affine coordinate for p; the sweep is injective iff no solution meets $\{\phi = 0,\ 0 \leq s \leq \alpha_2\}$ twice, and at every such meeting $\phi' = s - \alpha_2 \leq 0$, so the crossing is always downward. A second meeting therefore forces an upward crossing in between, at $s \geq \alpha_2$. Since the total rotation over $[0, \pi/2]$ is only $\pi/2$, the homogeneous part alone cannot supply that extra crossing; whether the forcing ζ can is the open point.

Proposition 30 (Monotonicity of the outer arm)**.** *The face-1 segment's far endpoint has x-coordinate*

$$x(t) = c_x(t) + \sigma(t) \sin t = \frac{F(t) - 1}{\cos t}, \tag{25}$$

the x-intercept of the face-1 line, and $\cos^2 t\, x'(t) = F'(t) \cos t + (F(t) - 1) \sin t$*. For* Σ*:*

- *on* $[0, \beta)$*, where* $A(t) \equiv P$ *so* $F(t) = \cos t + \frac{1}{2} \sin t$*,*

  $$x = 1 + \tfrac{1}{2} \tan t - \sec t, \qquad \cos^2 t\, x' = \tfrac{1}{2} - \sin t > 0,$$

  the last because $\sin \beta = 1/(4a_1) = 0.2856\ldots < \frac{1}{2}$ *by Theorem 14, i.e.* $\beta < \pi/6$*;*

- *on* $(\pi/2 - \beta, \pi/2]$*, where* $F(t) - 1 = A \cos t + \frac{1}{2} \sin t - \frac{1}{2}$ *with* $A = 1 - \frac{2}{3} a_1$*,*

  $$x = A - \tfrac{1}{2} \tan\Big(\tfrac{\pi}{4} - \tfrac{t}{2}\Big), \qquad \cos^2 t\, x' = \tfrac{1}{2}(1 - \sin t) > 0 \quad (t < \pi/2),$$

  and in particular $x(\pi/2) = A = 1 - \frac{2}{3} a_1 = 0.4164750917\ldots$*, which is exactly the right endpoint of the corridor floor facet of Remark 27;*

- *on* $[\beta, \pi/2-\beta]$, *substituting* $u = t/2 - \pi/8$ *and using* $-f_2/f_1 = \sqrt{2}-1 = \tan(\pi/8)$,

$$\cos^2 t\, x' = W(u) := \tfrac{3K}{4}\sin u + \tfrac{K}{4}\cos 3u + \tfrac{1}{2} - \sin\Big(2u + \tfrac{\pi}{4}\Big), \qquad K = f_1\sqrt{4-2\sqrt{2}},$$

on $|u| \le \pi/8 - \beta/2$.

Hence x *is strictly increasing, given* $W > 0$.

Proof. The face-1 line is $\{\langle p, \mu_t\rangle = F(t) - 1\}$, so its x-intercept is $(F-1)/\cos t$; that it agrees with $c_x + \sigma \sin t$ is $c_x + c_y \tan t = (F-1)/\cos t$, immediate from $c = (F-1)\mu_t + (G-1)\nu_t$. The three closed forms follow by substituting $F(t) - 1 = \mathrm{Re}(e^{-it}z(t))$ with the complex forms of SOL1, SOL6, SOL5. For the middle phase the κ_6 terms cancel and the triple-angle identities $\sin(3t/2) = s(3c^2 - s^2)$, $\cos(3t/2) = c(c^2 - 3s^2)$ with $s = \sin\frac{t}{2}$, $c = \cos\frac{t}{2}$ reduce the result to the stated three-term form. □

Proposition 31 ($W > 0$, by a finite certificate)**.** *$W > 0$ on $|u| \le u_0 := \pi/8 - \beta/2 = 0.2478721712\ldots$, so x is strictly increasing on all of $[0, \pi/2]$ and $S_1 = \sigma$.*

Proof. $K = 1.3020516916172893452\ldots$, and $3u_0 = 0.7436\ldots < \pi$, so $\cos 3u$ is unimodal on $[-u_0, u_0]$ with maximum at 0 and its minimum over any subinterval is attained at an endpoint. Also $\sin u$ is increasing, and $2u + \pi/4$ ranges over $[0.2896\ldots, 1.2812\ldots] \subset [0, \pi/2]$, so $\sin(2u+\pi/4)$ is increasing. Therefore on $[p, q] \subseteq [-u_0, u_0]$,

$$W \;\ge\; L(p,q) := \tfrac{3K}{4}\sin p + \tfrac{K}{4}\min\big(\cos 3p, \cos 3q\big) + \tfrac{1}{2} - \sin\Big(2q + \tfrac{\pi}{4}\Big).$$

Dividing $[-u_0, u_0]$ into 32 equal pieces gives $L > 0$ on every piece, with $\min L = 6.13 \cdot 10^{-3}$; the computation is at 40 working digits, so the margin exceeds the arithmetic error by 37 orders of magnitude. (16 pieces do not suffice: $\min L = -8.6 \cdot 10^{-3}$.) □

Remark 32. W is in fact strictly decreasing, with $W(u_0) = \frac{1}{2}(1 - \cos\beta) = 0.020828595599846555\ldots$ at the right junction, matching the last phase's $\frac{1}{2}(1 - \sin t)$ there to $1.7 \cdot 10^{-41}$, as C^1 gluing requires. The three closed forms of Proposition 30 were also checked against numerical differentiation of the path to 10^{-10}.

Remark 33 (The identity at 61 digits)**.** With all constants from their closed forms ($u^3 + 3u = 2$, $\beta = \arctan(u/2)$, $a_1 = \sqrt{4+u^2}/(4u)$, f_1 from Proposition 6) and 60 working digits,

$$V = 0.18419319708876898825165556 4606\ldots, \qquad |C_2| = 2.0133416126029788281721204768 13\ldots,$$

$$Q(\Sigma) - A_R^* = -1.56 \cdot 10^{-61}.$$

The two sides share no computation, so this rules out the possibility that the double-precision agreement of (23) was a rounding coincidence. It remains floating point, not an interval enclosure.

11 A convex subdomain on which the bound is valid

Proposition 28 says $V \geq |N|$ with equality exactly when nothing is re-swept, i.e. when the combined sweep is injective. We now show that injectivity is governed by inequalities that are *linear* in H, so the set where the bound is valid is convex.

Each sweep lies on a line: face 2 at parameter t on $\ell_2(t) = \{\langle p, \nu_t\rangle = G(t) - 1\}$, face 1 on $\ell_1(t) = \{\langle p, \mu_t\rangle = F(t) - 1\}$. Two distinct lines meet in one point, so a double cover forces that point into both segments. Solving $\ell_2(t) \cap \ell_2(t')$ for the coordinate s on $\ell_2(t)$, and using $\langle \mu_t, \nu_{t'}\rangle = \sin(t - t')$,

$$s_{22}(t, t') = \frac{\langle c(t), \nu_{t'}\rangle - (G(t') - 1)}{\sin(t - t')}, \qquad s_{11}(t, t') = \frac{\langle c(t), \mu_{t'}\rangle - (F(t') - 1)}{\sin(t' - t)}. \tag{26}$$

The numerator of s_{22} vanishes at $t' = t$ with derivative $-\alpha_2(t)$, so $s_{22}(t, t') \to \alpha_2(t)$: nearby face-2 lines meet at the *far* end of the face-2 segment $[0, \alpha_2]$. Likewise $s_{11}(t, t') \to \alpha_1(t)$, the *near* end of the face-1 segment $[\alpha_1^+, \sigma]$.

Definition 34. Let $\mathcal{K}'$ be the set of admissible H satisfying, for all $0 \leq t' < t \leq \pi/2$,

$$\text{(C22)} \quad \langle c(t), \nu_{t'}\rangle - (G(t') - 1) \ \geq \ \alpha_2(t)\sin(t - t'), \tag{27}$$

$$\text{(C11)} \quad \langle c(t), \mu_{t'}\rangle - (F(t') - 1) \ \leq \ \alpha_1(t)^+ \sin(t' - t) \ \ (\text{read with } \sin(t' - t) < 0), \tag{28}$$

together with the analogous condition for $\ell_2(t) \cap \ell_1(t')$.

Proposition 35. *$\mathcal{K}'$ is convex, and on $\mathcal{K}'$ the combined sweep is injective, hence $V = |N|$ and $Q = |C_2| - 2|N| \geq |T|$ for every ambidextrous sofa T with cap data H.*

Proof. By Lemma 18, $c(t)$ is affine in H, hence so are $\langle c(t), \nu_{t'}\rangle$, $G(t') - 1$, $F(t') - 1$, $\alpha_1(t)$ and $\alpha_2(t)$; $\sin(t - t')$ does not depend on H. So (27) and (28) are linear inequalities in H and cut out a convex set. Given (27), for any pair $t' < t$ the intersection point of $\ell_2(t)$ and $\ell_2(t')$ has $s_{22}(t, t') \geq \alpha_2(t)$, so it is not interior to the segment at t, and the pair contributes no double cover; (28) does the same for face 1 with the inequality reversed because the relevant endpoint is the near one. With the cross condition, no point is swept twice.

For $V = |N|$: every point of N has a first capture time $\tau(p)$, at which $\partial Q_{\tau(p)}$ is advancing at p (otherwise p would have been captured earlier), so by Lemma 19 it lies on face 2 with $s \leq \alpha_2$ or on face 1 with $s \geq \alpha_1$. Hence N is covered by the two sweeps, and the sweeps lie in N because an advancing face point belongs to $Q_{t+\varepsilon}$. Injectivity turns each $\int |\det|$ into the area of its image, and the two images are disjoint, so the sum is $|N|$. □

Remark 36 ($\Sigma \in \mathcal{K}'$, and $\mathcal{K}'$ has nonempty interior)**.** Measured on a grid of 401 parameters, Σ satisfies all three conditions: of the 158 802 ordered pairs with $|t - t'|$ above the diagonal cutoff, *zero* give a point interior to both face-2 segments and *zero* interior to both face-1 segments, and of 58 081 cross pairs, *zero* land in both. The margins in (27), (28) vanish linearly as $t' \to t$, which is forced: that is the envelope limit. Away from the diagonal they are strictly positive.

$\mathcal{K}'$ is not a single point. For $H = H_\Sigma + \varepsilon \sin(k\theta)$, which respects the gauge for even k, all conditions still hold with zero violations at $k = 2$ for $|\varepsilon| \leq 3 \cdot 10^{-2}$ and at $k = 4$ for $|\varepsilon| \leq 10^{-2}$, in both signs, and the two margins trade off smoothly. So Σ lies in $\mathcal{K}'$ with low-frequency room around it.

The mechanism of failure is identified: for the compactly supported bump perturbations of Remark 29, where $V - |N|$ was found positive, the face-2 condition continues to hold and it is the *face-1* condition that fails, by $-4.7 \cdot 10^{-2}$ at $\varepsilon = 0.05$ and $-2.3 \cdot 10^{-1}$ at $\varepsilon = 0.20$. Those perturbations have large higher derivatives, which is what creates face-1 envelope self-intersections.

Theorem 37 (Conditional)**.** *Assume:*

(i) $Q(\Sigma) = A_R^*$, $\delta Q(\Sigma) = 0$ *in every admissible direction, and* $\delta^2 Q \preceq 0$ *on the cell* $\mathcal{C}$ *of support functions sharing* Σ*'s sign pattern;*

(ii) the three conditions of Definition 34 hold on $\mathcal{K}' \cap \mathcal{C}$*;*

(iii) competitors satisfy the niche separation $M < \frac{1}{2}$ *of Corollary 2.*

Then $|T| \le A_R^*$ *for every ambidextrous sofa* T *whose cap data lies in* $\mathcal{K}' \cap \mathcal{C}$*.*

This statement is superseded *by Theorem 71, which reaches the same conclusion on an explicitly described convex domain with all three hypotheses proved. It is retained only because its three hypotheses isolate what the later argument has to supply, and because Section 12 closes hypothesis (ii) for it. Its own label is* HEURISTIC*; Theorem 71 is* PROVED*.*

Proof. $\mathcal{C}$ is convex because α_1, α_2 are affine in H, so each pointwise sign condition is a half-space; $\mathcal{K}'$ is convex by Proposition 35; hence so is the intersection. By (i), Q is a concave function on that convex set with a critical point at Σ, so $Q(H) \le Q(\Sigma) = A_R^*$ throughout. By (ii) and Proposition 35, $Q(H) \ge |T|$ there, using (iii) to get the factor 2 in $|T| \le |C_2| - 2|N|$. $\square$

12 One curvature condition implies injectivity

The conditions of Definition 34 were verified on a grid. We now prove them, for all pairs at once, from a single linear condition on the cap.

Definition 38. Say H satisfies (RC) if the absolutely continuous part of the surface measure obeys

$$H(\theta) + H''(\theta) \;\le\; 1 \qquad \text{for a.e. } \theta \in [0, \pi],$$

the only atoms being the corridor ceiling and floor facets at $\theta = \pm\pi/2$. Equivalently: the cap is nowhere flatter than a circle whose radius is the corridor width.

Since $H + H'' \le 1$ is linear in H, (RC) cuts out a convex set.

Remark 39 ((RC) is classical)**.** The bound $h + h'' \le R$ is not new. By the dual of Blaschke's rolling theorem, a convex body K slides freely inside a ball of radius R exactly when $h_K + h_K'' \le R$, and that in turn says K is a Minkowski summand of RB; see Schneider [4, §3.2]. Our (RC) is that condition at $R = 1$, the corridor width, *relaxed* to permit atoms at $\theta = \pm\pi/2$. Since Minkowski addition adds surface area measures, and a body whose surface measure is two equal atoms at $\pm\pi/2$ is a horizontal segment,

$$\text{(RC)} \quad \Longleftrightarrow \quad C_2 = \big[\text{a horizontal segment}\big] \oplus D, \qquad D \text{ a Minkowski summand of the unit disc.}$$

For Σ the segment has length $1.1670498\ldots$, the corridor facet, and D has curvature radius at most 0.8389. What we claim is not the condition but its use: that it forces the niche sweep to be injective (Theorems 40, 44).

Theorem 40. *If H satisfies (RC), then (27) and (28) hold for every pair $0 \leq t' < t \leq \pi/2$. Hence each of the two sweeps is injective.*

Proof. Fix t and set $\tau = t - t' \in (0, t]$. For (27) put $n(t') = \langle c(t), \nu_{t'} \rangle - (G(t') - 1)$ and

$$\Phi(\tau) = n(t - \tau) - \alpha_2(t) \sin \tau,$$

so that (27) reads $\Phi \geq 0$. Now $n(t) = 0$ because $\langle c(t), \nu_t \rangle = G(t) - 1$, and $n'(t') = -\langle c(t), \mu_{t'} \rangle - G'(t')$ equals $-(F(t) - 1) - G'(t) = -\alpha_2(t)$ at $t' = t$ by (18); hence $\Phi(0) = \Phi'(0) = 0$. Differentiating again and using $\nu'' = -\nu$,

$$n''(t') = -\langle c(t), \nu_{t'} \rangle - G''(t') = -\big(n(t') + G(t') - 1\big) - G''(t'),$$

so with $n(t - \tau) = \Phi(\tau) + \alpha_2(t) \sin \tau$ the α_2 terms cancel and

$$\Phi''(\tau) + \Phi(\tau) = 1 - (G + G'')(t - \tau) =: R(\tau). \tag{29}$$

With $\Phi(0) = \Phi'(0) = 0$, (29) gives $\Phi(\tau) = \int_0^\tau \sin(\tau - u) R(u)\, du$. Since $\tau \leq t \leq \pi/2 < \pi$ we have $\sin(\tau - u) \geq 0$ for $u \in [0, \tau]$, and $R \geq 0$ by (RC) because $G(s) = H(s + \pi/2)$. Therefore $\Phi \geq 0$.

The ceiling atom of H sits at $\theta = \pi/2$, i.e. at $t - u + \pi/2 = \pi/2$, i.e. $u = t$. For $\tau < t$ that is outside $[0, \tau]$, and for $\tau = t$ it is the endpoint, where the kernel $\sin(\tau - u)$ vanishes. So the atom contributes nothing.

For (28) put $m(t') = \langle c(t), \mu_{t'} \rangle - (F(t') - 1)$ and $\Psi(\tau) = m(t - \tau) + \alpha_1(t) \sin \tau$. Then $m(t) = 0$, $m'(t) = \alpha_1(t)$ by (17), so $\Psi(0) = \Psi'(0) = 0$, and $\mu'' = -\mu$ gives in the same way

$$\Psi''(\tau) + \Psi(\tau) = 1 - (F + F'')(t - \tau),$$

whence $\Psi(\tau) = \int_0^\tau \sin(\tau - u)\big[1 - (H + H'')(t - u)\big] du \geq 0$. Multiplying by $-\sin \tau < 0$ turns $\Psi \geq 0$ into $s_{11} \leq \alpha_1$, which is (28). Here the atom needs $t - u = \pi/2$, i.e. $u = t - \pi/2 \leq 0$, reachable only at $t = \pi/2$, where $\sigma(\pi/2) = \alpha_1(\pi/2)$ makes the face-1 segment empty and the condition vacuous. □

Remark 41 (Verification)**.** For Σ the absolutely continuous part of $H + H''$ has maximum $0.8385682\ldots$, at $\theta = \pi - \beta$, so (RC) holds with margin $0.1614\ldots$; the atom at $\theta = \pi/2$ has mass $1.1670498\ldots$, the ceiling facet length. The reduction (29) was checked against direct evaluation of $n(t - \tau) - \alpha_2(t) \sin \tau$ at several (t, τ), agreeing to $2.4 \cdot 10^{-5}$, the finite-difference floor for H''.

Theorem 40 also explains the failure mode of Remark 36. Across twelve perturbations, (RC) holds exactly when (28) holds: the compactly supported bumps give $\max(H + H'')_{\mathrm{ac}} =$ 1.08, 1.45, 2.18, 2.67, 4.58, 8.41 and all violate (28); $H_\Sigma + \varepsilon \sin(2\theta)$ at $|\varepsilon| = 0.03$ gives 0.888 and holds; $H_\Sigma + \varepsilon \sin(4\theta)$ gives 0.976 at $|\varepsilon| = 0.01$ and holds, 1.267 at $|\varepsilon| = 0.03$ and fails. The agreement is exact on both sides of the transition.

Remark 42 (What Theorem 40 does and does not give)**.** It replaces hypothesis (ii) of Theorem 37, for the two self-intersection conditions, by (RC): a single explicit linear inequality with a geometric reading. The cross condition, that $\ell_2(t) \cap \ell_1(t')$ avoid both segments, is not covered; it is measured to hold at Σ (0 of 58 081 pairs) and remains a hypothesis. So the hypothesis class of Theorem 37 may be replaced by $\{(\mathrm{RC})\} \cap \{\mathrm{cross}\} \cap \mathcal{C}$, an intersection of two convex sets with one measured condition.

Remark 43 (The first variation, pointwise)**.** Hypothesis (i) of Theorem 37 contains $\delta Q(\Sigma) = 0$. Integrating every η' in δQ by parts and collecting the coefficient of $\eta(\theta)$ gives the Euler–Lagrange equations of Q:

$$H + H'' = \alpha_2^+ + (\sigma - \alpha_1)\tan\theta - (\sigma - \alpha_1)' + (\alpha_1^-)' \qquad (0 < \theta < \pi/2),$$

$$H + H'' = -(\alpha_2^+)'(s) + \alpha_1^-(s), \qquad s = \theta - \pi/2 \qquad (\pi/2 < \theta < \pi).$$

Evaluated on Σ at eighteen values of θ spanning all six regions, the residual is at most $9.4 \cdot 10^{-6}$, which is the finite-difference floor. So $\delta Q(\Sigma) = 0$ has an explicit pointwise form; proving it amounts to substituting the closed forms of SOL1, SOL5, SOL6 phase by phase, and is not carried out here.

13 The cross condition, the first variation, and a Gårding bound

Theorem 44. *(RC) implies the cross condition: for all t, t' the point $\ell_2(t) \cap \ell_1(t')$ is not interior to both sweep segments. Consequently, under (RC) the combined sweep is injective and $V = |N|$, so $Q = |C_2| - 2|N| \geq |T|$.*

Proof. Work in the frame at t and write $p = c(t) + a\mu_t + b\nu_t$. The equation $\langle p, \nu_t\rangle = G(t) - 1$ forces $b = 0$, so p moves along $\ell_2(t)$ with $s_2 = -a$; the equation $\langle p, \mu_{t'}\rangle = F(t') - 1$ then gives, for $t' = t - \tau$,

$$s_2 = \frac{m(t - \tau)}{\cos\tau}, \qquad s_1 = -\Phi(\tau) - \big(\alpha_2(t) - s_2\big)\sin\tau,$$

using $\langle \mu_t, \nu_{t-\tau}\rangle = \sin\tau$ and $n(t - \tau) = \Phi(\tau) + \alpha_2(t)\sin\tau$. If $\tau > 0$ and p is interior to the face-2 segment, then $0 < s_2 < \alpha_2(t)$, and $\Phi \geq 0$ by Theorem 40 with $\sin\tau \geq 0$ because $\tau \leq \pi/2$; hence $s_1 < 0 \leq \alpha_1(t')^+$ and p lies below the face-1 segment.

For $t' > t$ work in the frame at t': the same computation with the roles of the two families exchanged gives, for $t = t' - \tau'$ with $\tau' > 0$,

$$s_2 = -\Psi(\tau') - \big(s_1 - \alpha_1(t')\big)\sin\tau',$$

so if p is interior to the face-1 segment then $s_1 > \alpha_1(t')^+ \geq \alpha_1(t')$ and $s_2 < 0$, outside $[0, \alpha_2(t)]$. At $t' = t$ the two lines are perpendicular and meet at $c(t)$, where $s_1 = s_2 = 0$, interior to neither. Both displayed identities were checked against a direct linear solve, agreeing to 10^{-16}. $\square$

So hypothesis (ii) of Theorem 37 may be replaced throughout by (RC).

Theorem 45. $\delta Q(\Sigma) = 0$. *Two things are needed and both hold: the Euler–Lagrange equations of Remark 43 hold identically on each of the six regions, as trigonometric identities in a_1, f_1, f_2 with no relation among the constants used; and the boundary term of δQ vanishes, because $H'(\pi) = -\frac{1}{2}$.*

Proof. On each phase, $F(t) - 1 = \mathrm{Re}(e^{-it}z(t))$ and $G(t) - 1 = \mathrm{Im}(e^{-it}z(t))$ evaluate to

SOL1	$F - 1 = \cos t + \frac{1}{2}\sin t - 1,$	$G - 1 = (2a_1 - 1)\sin t + \frac{1}{2}\cos t - \frac{1}{2},$
SOL6	$F - 1 = f_1 \cos\frac{t}{2} + f_2 \sin\frac{t}{2} - 1 + \kappa\cos t + \frac{1}{2}\sin t,$	$G - 1 = -f_2\cos\frac{t}{2} + f_1\sin\frac{t}{2} - 1 + \frac{1}{2}\cos t - \kappa\sin t,$
SOL5	$F - 1 = (1 - \frac{2}{3}a_1)\cos t + \frac{1}{2}\sin t - \frac{1}{2},$	$G - 1 = (\frac{8}{3}a_1 - 1)\sin t + \frac{1}{2}\cos t - 1,$

with $\kappa = 1-\frac{4}{3}a_1$. Substituting these into $\alpha_1 = G-1-F'$, $\alpha_2 = F-1+G'$, $\sigma = (F-1)\tan t+G-1$ and into the two Euler–Lagrange equations, with the sign pattern of each region fixed ($\alpha_2 > 0$ except on the last phase, $\alpha_1 < 0$ only on the first), every residual reduces identically to 0.

For the boundary term, integrating the η' of $\delta|C_2| = 2\int_0^\pi (H\eta - H'\eta')d\theta - \eta(0) - \eta(\pi)$ by parts and using $\eta(0) = 0$ leaves $-(2H'(\pi)+1)\eta(\pi)$ beyond the pointwise equations. The variation δV contributes nothing there: its only boundary term at $\theta = \pi$ carries the factor $\alpha_2^+(\pi/2)$, and $\alpha_2(\pi/2) < 0$, so the clamp kills it; every other boundary sits at $\theta = 0$ or $\pi/2$, where η vanishes. Finally the boundary point of C_2 with outer normal $\mu_\pi = (-1,0)$ is $H(\pi)\mu_\pi + H'(\pi)\nu_\pi = (-H(\pi), -H'(\pi))$, so its height is $-H'(\pi)$; since C_2 is ρ-symmetric about $y = \frac{1}{2}$ its leftmost point is ρ-fixed when unique, forcing $H'(\pi) = -\frac{1}{2}$ and the boundary term to vanish. The same mechanism at $\theta = 0$ gives $H'(0) = \frac{1}{2}$: both extreme points of the cap sit at height $\frac{1}{2}$. $\square$

Remark 46. The computation is a computer algebra verification, six residuals, each returning 0 with a_1, f_1, f_2 free. It was controlled against five deliberately perturbed variants of the middle-phase equation (sign flip on the tan term, each of the two terms dropped, α_1 and α_2 exchanged, $H + H''$ replaced by $H - H''$), all of which return nonzero; and the three pairs of phase formulas were checked against the reference path to $2.2\cdot 10^{-16}$. The content of Theorem 45 is that Romik's ODE families are exactly the critical-point equations of Q.

Remark 47 (Towards $\delta^2 Q \preceq 0$)**.** The remaining part of hypothesis (i) is the sign of the second variation. The step that makes it accessible is an exact algebraic cancellation. On $[0,\beta)$ write $p = \eta'(t)$, $q = \eta(t)$, $r = \eta(t+\pi/2)$, $T = \tan t$; the three terms of (24) that carry $\eta'(t)$ combine as

$$-p^2 - (qT+p)^2 + (r-p)^2 = -(p+qT+r)^2 + 2r^2 + 2qTr, \tag{30}$$

so the added term is absorbed exactly and the remainder carries no derivative of η. What is left is a lower-order estimate. With $\eta(0) = \eta(\pi/2) = 0$ the relevant Poincaré constants are 4 on $[0,\pi/2]$, 1 on $[\pi/2,\pi]$ (Dirichlet–Neumann, $\eta(\pi)$ free) and $(\pi/2\beta)^2 = 29.41$ on $[0,\beta]$. Retaining a fraction δ of $-\int_0^\beta \eta'^2$ before applying (30) and splitting $(qT+r)^2$ with a parameter κ, the coefficient of $\int_0^\beta \eta^2$ is

$$\left(\tfrac{1+\kappa}{1-\delta} - 1\right)\tan^2\beta + 1 - \delta\left(\tfrac{\pi}{2\beta}\right)^2 = -1.83\ldots < 0 \qquad (\delta = \tfrac{1}{10},\ \kappa = 1),$$

and the r^2 remainder, of coefficient $3.22\ldots$, is absorbed by the E_2 term of (24), which supplies $-\frac{1}{2}\int_{\pi/2}^{\pi-\beta} \eta'^2$: since $\eta(\pi/2) = 0$ gives $\int_{\pi/2}^{\pi/2+\beta} \eta^2 \le (\beta^2/2)\int \eta'^2$, the requirement is $3.22 \cdot 0.0420 = 0.135 \le \frac{1}{2}$. So every ingredient of a Gårding inequality $\delta^2 Q \le -c\|\eta\|_{L^2}^2$ is in place; Theorem 48 assembles it and Theorem 72 sharpens the constant to $\frac{2}{3}$.

The sharp value is $\frac{1}{2}c = 0.7309566\ldots$ by Theorem 73. A P1 finite-element computation approaches it from above, as Rayleigh–Ritz must: $-0.735818, -0.733924, -0.733086, -0.732285, -0.731311, -0.7312$ at $m = 32, \ldots, 2048$.

14 The area identity

The concavity of Q is established in Section 20; what follows here is the earlier and weaker route to the constant, retained for its elementary ingredients, and the closed-form evaluation of $Q(\Sigma)$.

Remark 48 (An earlier route to the constant)**.** Superseded by Theorem 73, which gives the exact constant, and by Theorem 76, which gives concavity on the whole ordered anchored family. It is retained because its ingredients are elementary and its failure locates the loss. On the cell $\mathcal{C}$ of Σ's sign pattern, $\delta^2 Q \le 0$, with equality only at $\eta = 0$. Explicitly, writing $J_1 = \int_0^\beta \eta^2$, $J_2 = \int_\beta^{\pi/2} \eta^2$, $J_3 = \int_{\pi/2}^{\pi} \eta^2$,

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -1.6013\,J_1 - 0.3710\,J_2 + 0\cdot J_3 - 0.005\int_{\pi/2}^{\pi-\beta} \eta'^2. \tag{31}$$

Proof of the statement in Remark 48. On $[0,\beta)$ retain a fraction δ of $-\int_0^\beta \eta'^2$ and apply (30) with the remaining $-(1-\delta)\eta'^2$; completing the square in $p = \eta'(t)$ and discarding the resulting negative square,

$$-(1-\delta)p^2 - (qT+p)^2 + (r-p)^2 \;\le\; \frac{(qT+r)^2}{1-\delta} - q^2T^2 + r^2 \;\le\; A\,q^2T^2 + B\,r^2,$$

with $A = \frac{1+\kappa}{1-\delta} - 1$, $B = \frac{1+1/\kappa}{1-\delta} + 1$ from $(qT+r)^2 \le (1+\kappa)q^2T^2 + (1+1/\kappa)r^2$, and $T = \tan t \le \tan\beta$ on $[0,\beta)$. The three Poincaré constants for $\eta(0) = \eta(\pi/2) = 0$ are $P_1 = (\pi/2\beta)^2 = 29.4135$ on $[0,\beta]$, $P_2 = (\pi/2(\pi/2-\beta))^2 = 1.5035$ on $[\beta,\pi/2]$ and $P_3 = 1$ on $[\pi/2,\pi]$, each with a Dirichlet endpoint and a free one. Finally $-\big(\eta(t) + \eta'(t+\pi/2)\big)^2 \le -\lambda\,\eta'(t+\pi/2)^2 + \frac{\lambda}{1-\lambda}\eta(t)^2$ turns the E_2 term into $-\lambda\int_{\pi/2}^{\pi-\beta} \eta'^2 + \frac{\lambda}{1-\lambda}(J_1+J_2)$, and $\eta(\pi/2) = 0$ gives $\int_{\pi/2}^{\pi/2+\beta} \eta^2 \le (\beta^2/2)\int_{\pi/2}^{\pi-\beta} \eta'^2$, so the $B\,r^2$ remainder is absorbed as soon as $\lambda \ge B\beta^2/2$. Collecting,

$$\text{coefficient of } J_1 = A\tan^2\beta - \delta P_1 + 1 + \tfrac{\lambda}{1-\lambda}, \qquad \text{of } J_2 = 1 - P_2 + \tfrac{\lambda}{1-\lambda},$$

$$\text{of } J_3 = 1 - P_3 = 0, \qquad \text{of } \int_{\pi/2}^{\pi-\beta} \eta'^2 = B\beta^2/2 - \lambda.$$

At $\delta = \frac{1}{10}$, $\kappa = 2$, $\lambda = 0.117$ these are -1.6013, -0.3710, 0 and -0.005, giving (31). For strictness, equality forces $\eta \equiv 0$ on $[0,\pi/2]$ and η the first Dirichlet–Neumann mode on $[\pi/2,\pi]$; but with $\eta \equiv 0$ on $[0,\pi/2]$ the E_2 term contributes $-\int_0^{\pi/2-\beta} \eta'(t+\pi/2)^2 < 0$. □

The bound $\frac{1}{2}\delta^2 Q \le -0.3710\|\eta\|^2_{L^2(0,\pi/2)}$ that (31) yields is about half of the sharp $-0.7309566\ldots$ for the full interval, the shortfall being the J_3 coefficient that (31) leaves at zero. Theorems 72 and 73 recover it.

Proposition 49. *$\sigma - \alpha_1 = \cos t\, x'(t)$ on each phase, with $x = (F-1)/\cos t$ as in (25). Hence the middle term of (20) is $\frac{1}{2}\int \cos^2 t\, x'(t)^2\, dt$, an energy of the intercept path.*

Proof. $\sigma - \alpha_1 = (F-1)\tan t + F'$ by (17) and (19), and $\cos^2 t\, x' = F'\cos t + (F-1)\sin t$ by (25); dividing by $\cos t$ gives the claim. Verified symbolically on SOL1, SOL5 and SOL6, residual 0 in each case, which also recovers the first and third closed forms of Proposition 30. □

Remark 50 (The area identity)**.** Since $u = 2\tan\beta$, Romik's area is

$$A_R^* = u^2 + 1 + \arctan(u/2) = 1 + 4\tan^2\beta + \beta.$$

Using Proposition 49 to make every integrand elementary, the six blocks of $|C_2|$ and the three blocks of V evaluate at 50 working digits to $|C_2| = 2.0133416126029788281721\ldots$, $V = 0.1841931970887689882517\ldots$, and

$$Q(\Sigma) - \big(1 + 4\tan^2\beta + \beta\big) = -2.7\cdot 10^{-51}.$$

A symbolic evaluation of the same integrals was attempted and did not terminate; the identity therefore remains HEURISTIC, now against a closed-form target and with a phase-by-phase decomposition.

15 Σ satisfies (RC), and $Q(\Sigma) = A_R^*$

Proposition 51. *Σ satisfies (RC). The surface-measure density is, block by block,*

$$H+H'' = \begin{cases} 0 & \theta \in [0,\beta) \text{ and } \theta \in (\pi-\beta, \pi], \\ \frac{1}{2} & \theta \in (\pi/2-\beta, \pi/2) \text{ and } \theta \in (\pi/2, \pi/2+\beta), \\ \frac{3K}{4}\cos\left(\frac{\theta}{2}+\frac{\pi}{8}\right) & \theta \in [\beta, \pi/2-\beta], \\ \frac{3K}{4}\sin\left(\frac{\theta-\pi/2}{2}+\frac{\pi}{8}\right) & \theta \in [\pi/2+\beta, \pi-\beta], \end{cases} \qquad K = f_1\sqrt{4-2\sqrt{2}},$$

plus the ceiling atom at $\theta = \pi/2$. Hence

$$\max(H+H'')_{\mathrm{ac}} = \tfrac{3K}{4}\cos\left(\tfrac{\beta}{2}+\tfrac{\pi}{8}\right) = 0.8388253494\ldots < 1,$$

and already the amplitude satisfies $\frac{3}{4}K = 0.9765387687\ldots < 1$, so (RC) holds as soon as

$$f_1^2 \ \le\ \frac{16}{9\,(4-2\sqrt{2})} = 1.5174\ldots, \qquad \textit{against } f_1^2 = 1.4470\ldots \tag{32}$$

Proof. Each block of H is read off the three closed forms of Theorem 45. On the first phase $H = \cos\theta + \frac{1}{2}\sin\theta$, so $H + H'' = 0$; the shifted last phase is the same. On the last phase $H = (1-\frac{2}{3}a_1)\cos\theta + \frac{1}{2}\sin\theta + \frac{1}{2}$ and on the shifted first phase $H = (2a_1-1)\sin\theta + \frac{1}{2}\cos\theta + \frac{1}{2}$, each giving $H + H'' = \frac{1}{2}$. On the middle phase only the half-angle terms survive differentiation twice, contributing a factor $1 - \frac{1}{4}$, so $H + H'' = \frac{3}{4}(f_1\cos\frac{\theta}{2} + f_2\sin\frac{\theta}{2})$ and, on the shifted middle phase, $\frac{3}{4}(f_1\sin\frac{\theta}{2} - f_2\cos\frac{\theta}{2})$. Both have amplitude $\frac{3}{4}\sqrt{f_1^2+f_2^2} = \frac{3}{4}f_1\sqrt{4-2\sqrt{2}}$ by $f_2 = (1-\sqrt{2})f_1$, and phase $\pi/8$ because $-f_2/f_1 = \sqrt{2}-1 = \tan(\pi/8)$. On $[\beta, \pi/2-\beta]$ the argument $\frac{\theta}{2}+\frac{\pi}{8}$ runs over $[0.5375, 1.0333] \subset [0, \pi/2]$, where cosine decreases, so the maximum is at $\theta = \beta$. □

Criterion (32) is strictly stronger than the separation criterion $f_1^2 < (2+\sqrt{2})/2 = 1.7071\ldots$ of Theorem 8: within the family in which f_1 is the free parameter, (RC) implies $M < \frac{1}{2}$.

Theorem 52. $Q(\Sigma) = A_R^*$.

Proof. Σ satisfies (RC) by Proposition 51, so $V(\Sigma) = |N(\Sigma)|$ by Theorems 40 and 44. The niches are disjoint by Theorem 8, and C_2 is ρ-invariant, so $|\Sigma| = |C_2| - |N| - |\rho N| = |C_2| - 2|N|$. Hence $Q(\Sigma) = |C_2| - 2V(\Sigma) = |C_2| - 2|N| = |\Sigma| = A_R^*$. □

No integral is evaluated. Independently, with $A_R^* = 1 + 4\tan^2\beta + \beta$ and the integrands made elementary by Proposition 49, the two sides agree to $2.7 \cdot 10^{-51}$ at 50 working digits, which is a consistency check on the whole chain rather than a proof of it.

Remark 53 (Regression against the one-corner problem)**.** The construction does *not* recover Gerver's sofa, and this should be stated plainly. Applying the same machinery to Gerver's rotation path:

- Gerver satisfies the one-corner injectivity condition everywhere: $\alpha_1 > 0$ and $\alpha_2 > 0$ on all of $[0, \pi/2]$, as expected.

- (RC) *fails* for Gerver's cap: $\max(H + H'')_{\mathrm{ac}} = 1.3960$ near $\theta = \pi$.
- Gerver's face-2 sweep is genuinely not injective in this decomposition: 33 002 of 158 802 tested pairs meet interior to both segments, with $\min(s_{22} - \alpha_2) = -0.0944$. The face-1 sweep is injective.
- Consistently, $V = 0.6426146$ against $|N| = |C| - |S| = 0.6406$ to 0.6412 at 481 and 721 constraints, i.e. $V > |N|$ as Proposition 28 requires.

So (RC) and the one-corner injectivity condition are independent: Σ satisfies the first and not the second, Gerver the second and not the first. By Remark 39 the difference has a classical reading: Σ's cap is a corridor facet plus a body that slides freely inside the unit disc, whereas part of Gerver's cap boundary is *flatter than the corridor is wide*, its curvature radius reaching 1.3986. The decomposition $N = W_2 \sqcup W_1^{\mathrm{out}}$ is therefore not the one the one-corner argument uses, and the fact that it fails on the solved case is a standing caution about the framework, recorded rather than explained.

Remark 54 (The separation hypothesis for competitors)**.** Hypothesis (iii) remains open. Over 250 random perturbations $H_\Sigma + \sum_{k=1}^{3} c_k \sin(2k\theta)$ with $|c_k| \le a/k^2$ for $a = 0.02, 0.05, 0.10$, every H satisfying (RC) had $M \le 0.3976$, against the required $M < \frac{1}{2}$; the (RC)-satisfying fraction fell from 250/250 to 34/250 as a grew. No counterexample and no implication.

16 Two negative findings

Remark 55 (The mirror decomposition does not rescue Gerver)**.** The decomposition of Proposition 20 has a mirror image under $t \mapsto \pi/2 - t$, $\mu \leftrightarrow \nu$: sweep the inner part of face 1 and the outer part of face 2, the latter reaching the floor at distance $\sigma_2 = c_y/\sin t$. For Gerver both give the same value, $0.6426144\ldots$, because Gerver's rotation path is itself symmetric under $t \mapsto \pi/2 - t$; both exceed $|N| \approx 0.641$. So the failure of Remark 53 is not an artefact of orientation.

The structural reason is visible in Proposition 51. Σ's cap has $H + H'' = 0$ on $[0, \beta)$ and $(\pi - \beta, \pi]$, i.e. *vertices*, because Σ's outer phases have constant contact arcs pinned at $(1, \frac{1}{2})$. Gerver's cap has no such vertices: there $H + H''$ reaches 1.3960 near $\theta = \pi$. The degeneracy that makes Σ hard for local analysis is exactly what makes (RC) hold for it.

Remark 56 (A false bound, retracted)**.** The following was attempted for hypothesis (iii) and is **false**. Solving $H'' + H = W$ with $H(0) = 1$ gives $H(\theta) = \cos\theta + H'(0)\sin\theta + \int_0^\theta \sin(\theta - s)W(s)\,ds$, and substituting into $c_y(t) = (H(t) - 1)\sin t + (H(t + \pi/2) - 1)\cos t$ yields

$$c_y(t) = H'(0) - \sin t - \cos t + \sin t \int_0^t \sin(t - s)W\,ds + \cos t \int_0^{t+\pi/2} \cos(t - s)W\,ds.$$

Both kernels are non-negative on their ranges, so "$W \le 1$" would give $c_y \le H'(0)$, and for a ρ-symmetric cap $H'(0) = \frac{1}{2} + \frac{1}{2}$(vertical right edge), apparently yielding $M \le \frac{1}{2}$.

The error is that W is *not* ≤ 1 as a measure: it carries the ceiling atom of mass $1.1670\ldots$ at $s = \pi/2$, and here that atom meets the kernel $\cos(t - \pi/2) = \sin t$, which does not vanish. The corrected bound is $M \le H'(0) + \frac{1}{2} \cdot 1.1670 = 1.083\ldots$, useless against the required $\frac{1}{2}$.

This does *not* affect Theorem 40: there the same atom lands at $u = t$, which is outside the integration range when $\tau < t$ and is the endpoint where the kernel $\sin(\tau - u)$ vanishes when

$\tau = t$; and the face-1 atom needs $u = t - \pi/2 \le 0$. The placement of the atom relative to the kernel is what distinguishes the two arguments, and it was checked in both.

Hypothesis (iii) therefore remains open.

17 A finite-angle relaxation, and what it can give

Everything above is local. We record here the one construction in this note that bears on the *global* problem, together with an honest account of how far it has been carried.

Proposition 57 (Finite-angle relaxation)**.** *Let S be an ambidextrous moving sofa. For any finite $F_1, F_2 \subset [0, \pi/2]$,*

$$|S| \;\le\; \sup\Big|\bigcap_{t\in F_1} L(t, c_t) \;\cap\; \bigcap_{s\in F_2} R(s, d_s)\Big|,$$

the supremum being over all placements $c_t, d_s \in \mathbb{R}^2$, where $L(t, c)$ is the left-turning hallway rotated by t with inner corner c, and $R(s, d)$ the right-turning one, that is, the mirror image rotated by s and placed at d.

Proof. S lies in the hallway at every instant of each of its two motions, so it lies in the intersection over any finite subset of instants, for the placements those instants realise. Dropping the constraints linking the placements to one another enlarges the right-hand side. □

The bound this yields depends on F_1, F_2 and requires the supremum, which is a global maximisation over $2(|F_1| + |F_2|) - 2$ real parameters. Two facts are recorded.

Proposition 58 (A limitation at three angles)**.** *For $F_1 = F_2 = \{0, \pi/4, \pi/2\}$ the supremum in Proposition 57 is at least $2\sqrt{2} - 1 = 1.8284271\ldots$, so no bound below that value is obtainable at those angles.*

Proof. An explicit placement attaining it is recorded in the accompanying computation; exhibiting one placement suffices. □

Three objects already force this value, the surrounding structure is classical, and the constant has an exact geometric meaning.

A moving sofa lies in the incoming straight corridor before it reaches the corner, so $S \subseteq \mathcal{S}$ for a strip $\mathcal{S}$ of width 1; an ambidextrous one meets a $\pi/4$ hallway of each handedness. Hence

$$|S| \;\le\; \sup_{c,d} \big|\mathcal{S} \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\big|. \tag{33}$$

With only $L_{\pi/4}$ this is Hammersley's bound: each arm of that hallway is a strip of width 1 at $\pi/4$ to $\mathcal{S}$, meeting it in a parallelogram of area $1/\sin(\pi/4) = \sqrt{2}$, so $|S| \le 2\sqrt{2}$. The second handedness changes the geometry, and the following is what the constant $2\sqrt{2} - 1$ measures.

Lemma 59 (Cap lemma)**.** *Let T be a unit square whose diagonals are horizontal and vertical, and $\mathcal{S}$ a horizontal strip of width 1. Then $|T \cap \mathcal{S}| \le \sqrt{2} - \frac{1}{2}$, with equality when the strip is centred on T.*

Proof. T has vertical extent $\sqrt{2} > 1$, and its horizontal cross-section at height y from the centre has length $\sqrt{2} - 2|y|$ for $|y| \le \sqrt{2}/2$. The cap above height h has area $\int_h^{\sqrt{2}/2} (\sqrt{2} - 2y)\, dy = \frac{1}{2} - \sqrt{2}\, h + h^2$, decreasing in h on $[0, \sqrt{2}/2]$, so a strip of width 1 removes least when centred, at $h = \frac{1}{2}$:

$$|T \cap \mathcal{S}| \;\le\; 1 - 2\big(\tfrac{1}{2} - \tfrac{\sqrt{2}}{2} + \tfrac{1}{4}\big) = \sqrt{2} - \tfrac{1}{2}. \qquad \square$$

All four arms of the two $\pi/4$ hallways are strips of width 1 with normal $(1,1)/\sqrt{2}$ or $(-1,1)/\sqrt{2}$. Two arms from different hallways with perpendicular normals meet in exactly such a square, so each perpendicular pair contributes at most $\sqrt{2} - \frac{1}{2}$ by Lemma 59, and

$$2\big(\sqrt{2} - \tfrac{1}{2}\big) = 2\sqrt{2} - 1.$$

This is not merely numerology: the maximising configuration found for (33) splits into exactly two connected components, each of area $0.914213562 = \sqrt{2} - \frac{1}{2}$, one for each perpendicular pair, with the two parallel pairs empty. The corners sit at heights 0.995169 and 0.004831, symmetric about the midline.

The conjecture below reduces to a one-dimensional problem. Let $\mu = (1,1)/\sqrt{2}$, $\nu = (-1,1)/\sqrt{2}$ and use the orthonormal coordinates $u = \langle p, \mu\rangle$, $v = \langle p, \nu\rangle$; write $a = \langle c, \mu\rangle$, $a' = \langle c, \nu\rangle$, $b = \langle d, \mu\rangle$, $b' = \langle d, \nu\rangle$.

Lemma 60 (Rotated coordinates)**.** *In these coordinates both hallways are axis-aligned and*

$$L_{\pi/4}(c) \cap R_{\pi/4}(d) = \big([b-1, a+1] \times [b'-1, a'+1]\big) \setminus \big(\{u < a,\ v < a'\} \cup \{u > b,\ v > b'\}\big),$$

a rectangle less two opposite quadrants, while $\mathcal{S} = \{0 \le u + v \le \sqrt{2}\}$. Consequently, with $s = u + v$ and $t = u - v$,

$$\big|\mathcal{S} \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\big| = \tfrac{1}{2} \int_0^{\sqrt{2}} \ell(s)\, ds, \tag{34}$$

where $\ell(s)$ is the length of $\big[\max(p, r) - 2,\ \min(q, w) + 2\big] \setminus \big((p, q) \cup (r, w)\big)$ with $p = s - 2a'$, $q = 2a - s$, $r = 2b - s$, $w = s - 2b'$. In particular ℓ is piecewise linear in s, and the two removed intervals are nonempty exactly for $s < a + a'$ and $s > b + b'$ respectively.

Proof. $L_{\pi/4}(c) = \{u \le a + 1, v \le a' + 1\} \setminus \{u < a, v < a'\}$ and $R_{\pi/4}(d) = \{u \ge b - 1, v \ge b' - 1\} \setminus \{u > b, v > b'\}$ directly from the definitions, since reflection carries μ, ν to $-\nu, -\mu$. The cross-section of the rectangle at level s is $[\max(p, r) - 2, \min(q, w) + 2]$ in t, the two quadrants cut out (p, q) and (r, w), and $du\, dv = \frac{1}{2}\, ds\, dt$. $\square$

Formula (34) agrees with direct polygon computation to $2.5 \cdot 10^{-11}$ on test placements. The crude bound $\ell \le 4$ recovers Hammersley: it gives $\frac{1}{2} \cdot 4\sqrt{2} = 2\sqrt{2}$. The conjectured value is attained, and can be computed exactly.

Proposition 61 (The extremal configuration)**.** *At $a = a' = b = b' = \sqrt{2}/4$, that is with both corners at the same point at mid-corridor height, $\ell(s) = 4 - 4\big|s - \frac{\sqrt{2}}{2}\big|$ and (34) equals exactly $2\sqrt{2} - 1$.*

Proof. Put $\delta = s - \sqrt{2}/2$. Then $p = w = \delta$ and $q = r = -\delta$, so the outer interval is $[|\delta| - 2,\ 2 - |\delta|]$, of length $4 - 2|\delta|$. Exactly one of $(p, q) = (\delta, -\delta)$ and $(r, w) = (-\delta, \delta)$ is nonempty, according to the sign of δ, and it has length $2|\delta|$ and lies inside the outer interval because $|\delta| \le \sqrt{2}/2 < 1$. Hence $\ell = 4 - 4|\delta|$ and $\frac{1}{2} \int_0^{\sqrt{2}} \big(4 - 4|\delta|\big) ds = \frac{1}{2}\big(4\sqrt{2} - 4 \cdot \frac{1}{2}\big) = 2\sqrt{2} - 1$. $\square$

A search of 600 multistarts over $[-3,3]^4$ in exact piecewise-linear arithmetic returns 1.828427124746, agreeing with $2\sqrt{2}-1$ to $2.7\cdot 10^{-15}$, and no configuration exceeding it was found.

The supremum in (33) can be evaluated in closed form on most of the parameter space, and this gives the first upper bound specific to the ambidextrous problem. The key is that the set left after the two quadrants are removed is confined to two short end pieces, whose lengths are controlled by two gauge-invariant combinations of c and d.

Lemma 62 (Four pieces)**.** *With the notation of Lemma 60, set*

$$m_1 = \langle c,\nu\rangle + \langle d,\mu\rangle = a' + b, \qquad m_2 = \langle c,\mu\rangle + \langle d,\nu\rangle = a + b',$$

$$\delta_\mu = \langle d-c,\mu\rangle = b-a, \qquad \delta_\nu = \langle d-c,\nu\rangle = b'-a'.$$

All four are unchanged by the translation $(a,a',b,b')\mapsto(a+\varepsilon,a'-\varepsilon,b+\varepsilon,b'-\varepsilon)$, and for every s

$$\ell(s) \;\le\; \bigl(2-2|s-m_1|\bigr)^+ + \bigl(2-2|s-m_2|\bigr)^+ + \min\bigl(2(s-m_1)^+, 2\delta_\mu^+\bigr) + \min\bigl(2(s-m_2)^+, 2\delta_\nu^+\bigr). \quad (35)$$

Proof. Write $I = [\max(p,r)-2, \min(q,w)+2]$. Since $I\setminus(p,q) = \{x\in I : x\le p\}\cup\{x\in I : x\ge q\}$ and likewise for (r,w), intersecting the two gives

$$I\setminus\bigl((p,q)\cup(r,w)\bigr) \;=\; \bigl(A\cup B\cup C\cup D\bigr)\cap I,$$

$$A = (-\infty,\min(p,r)], \quad B=[q,r], \quad C=[w,p], \quad D=[\max(q,w),\infty).$$

Now $|A\cap I|\le \min(p,r)-\max(p,r)+2 = (2-|p-r|)^+$ and $p-r = 2s-2(a'+b)$, giving the first term; $|D\cap I|\le\min(q,w)+2-\max(q,w) = (2-|q-w|)^+$ and $q-w = 2(a+b')-2s$, giving the second. For the cross piece $C=[w,p]$: it is empty unless $p>w$, i.e. $\delta_\nu>0$, and $C\setminus A\subseteq[\min(p,r),p]$, so $|C\setminus A|\le\min\bigl((p-r)^+,(p-w)^+\bigr) = \min\bigl(2(s-m_1)^+, 2\delta_\nu^+\bigr)$. Symmetrically B is empty unless $\delta_\mu>0$ and $|B\setminus D|\le\min\bigl((w-q)^+,(r-q)^+\bigr) = \min\bigl(2(s-m_2)^+,2\delta_\mu^+\bigr)$. Adding the four bounds can only over-count overlaps. □

Proposition 63 (The profile integral)**.** *Let $g(x)=\int_0^{\sqrt{2}}\bigl(2-2|s-x|\bigr)^+ds$. Then $g(\sqrt{2}-x)=g(x)$, g vanishes off $(-1,\sqrt{2}+1)$, on $[\sqrt{2}-1,1]$ it equals $2\sqrt{2}-x^2-(\sqrt{2}-x)^2$, and*

$$\max_{x\in\mathbb{R}} g(x) \;=\; g\bigl(\tfrac{1}{\sqrt{2}}\bigr) \;=\; 2\sqrt{2}-1.$$

Proof. The substitution $s\mapsto\sqrt{2}-s$ gives the symmetry. Differentiating under the integral, $g'(x) = 2\bigl|[0,\sqrt{2}]\cap(x-1,x)\bigr| - 2\bigl|[0,\sqrt{2}]\cap(x,x+1)\bigr|$, which is ≤ 0 for $x\ge\sqrt{2}/2$ because the second window then contains at least as much of $[0,\sqrt{2}]$ as the first; with the symmetry, g increases up to $\sqrt{2}/2$ and decreases after. For $\sqrt{2}-1\le x\le 1$ the integrand is nonnegative throughout $[0,\sqrt{2}]$, so $g(x) = 2\sqrt{2}-2\int_0^{\sqrt{2}}|s-x|\,ds = 2\sqrt{2}-x^2-(\sqrt{2}-x)^2$, and at $x=\sqrt{2}/2$ this is $2\sqrt{2}-\frac{1}{2}-\frac{1}{2}$. □

Combining (35) with $\ell\ge 0$ and integrating over the band gives a bound in closed form, with no computation of any kind.

Theorem 64 (The sharp constant, off the cross region)**.** *For all c, d,*

$$\big|\mathcal{C}_2 \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\big| \;\le\; \tfrac{1}{2}\big[g(m_1)+g(m_2)\big] + \sqrt{2}\big[\min(1,\delta_\mu^+)+\min(1,\delta_\nu^+)\big]. \tag{36}$$

In particular, if $\langle d-c,\mu\rangle \le 0$ and $\langle d-c,\nu\rangle \le 0$ then

$$\big|\mathcal{C}_2 \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\big| \;\le\; 2\sqrt{2}-1 \;=\; 1.8284271\ldots,$$

with equality exactly when $m_1 = m_2 = 1/\sqrt{2}$ and $\delta_\mu = \delta_\nu = 0$. Consequently every ambidextrous moving sofa whose two corner positions at angle $\pi/4$ satisfy those two inequalities has area at most $2\sqrt{2}-1$.

Proof. Integrate (35) over $s \in [0,\sqrt{2}]$ and halve, as in (34). The first two terms give $\frac{1}{2}g(m_1) + \frac{1}{2}g(m_2)$ by definition of g; the third is at most $\frac{1}{2}\int_0^{\sqrt{2}} 2\min(1,\delta_\mu^+)\,ds = \sqrt{2}\min(1,\delta_\mu^+)$ and likewise the fourth. If $\delta_\mu \le 0$ and $\delta_\nu \le 0$ the cross terms are absent and Proposition 63 bounds the rest by $\frac{1}{2}\cdot 2(2\sqrt{2}-1)$. Equality forces both g's to be maximal, hence $m_1 = m_2 = 1/\sqrt{2}$; the configuration $a = a' = b = b' = \sqrt{2}/4$ of Proposition 61 realises it, and (36) is therefore sharp. □

The bound (36) was tested against the objective computed independently from the definition at $4\cdot10^4$ random parameter triples over scales from 0.1 to 160: the minimum slack is $-1.0\cdot10^{-14}$, i.e. zero to rounding, so the bound is both correct and attained.

When $\delta_\mu > 0$ or $\delta_\nu > 0$ the cross terms in (36) are genuinely present and the estimate exceeds $2\sqrt{2}-1$. The next three lemmas remove the sign hypothesis. Throughout, the reflection across the diagonal $u = v$ preserves the band and exchanges (a,a') with (a',a) and (b,b') with (b',b), hence $\delta_\mu \leftrightarrow \delta_\nu$ and $m_1 \leftrightarrow m_2$; so assume without loss of generality $t := \delta_\nu = \max(\delta_\mu,\delta_\nu) > 0$.

Lemma 65 (Wide offsets)**.** *If $t \ge 1$ then $\big|\mathcal{C}_2 \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)\big| \le \sqrt{2}\,(2-t)^+ \le \sqrt{2}$.*

Proof. $w - p = (s-2b') - (s-2a') = -2t$ identically in s, so $\ell(s) \le |I| \le 4 + w - p = 4 - 2t$ for every s, and the band has length $\sqrt{2}$: $\frac{1}{2}\int_0^{\sqrt{2}} \ell \le \frac{1}{2}\sqrt{2}\,(4-2t)^+$. □

Lemma 66 (Both offsets positive)**.** *If $0 < \delta_\mu \le \delta_\nu = t \le 1$ then $\ell(s) \le (2-2|s-m_1|)^+ + (2-2|s-m_2|)^+$ for every s, so the area is at most $\frac{1}{2}[g(m_1)+g(m_2)] \le 2\sqrt{2}-1$.*

Proof. Write, as in Lemma 62, the kept set as $(A \cup B \cup C \cup D) \cap I$ with $A = (-\infty, \min(p,r)]$, $B = [q,r]$, $C = [w,p]$, $D = [\max(q,w),\infty)$, and note $m_1 \le m_2$ since $\delta_\mu \le \delta_\nu$. For $s \le m_1$ we have $p \le r$ and $w \le q$, so $C \subseteq A$ and $B \subseteq D$, and $\ell \le |A\cap I| + |D \cap I| \le (2-2(m_1-s))^+ + (2-2(m_2-s))^+$ exactly as in the proof of Theorem 64; symmetrically for $s \ge m_2$. On the middle window $m_1 < s < m_2$: $A = (-\infty, r]$, $D = [q,\infty)$, and since $\delta_\mu > 0$ gives $q \le r$,

$$A \cap D = [q,r] = B, \qquad A \cap C \cap D = [q,r] \cap [w,p] = [q,r] = B,$$

the latter because $w \le q$ and $r \le p$ on the window. In the inclusion–exclusion for $|A \cup C \cup D|$ the terms $-|A\cap D|$ and $+|A \cap C \cap D|$ therefore cancel, leaving

$$\ell = |A\cap I| + |D\cap I| + \big[\,|C\cap I| - |A \cap C \cap I| - |C \cap D \cap I|\,\big].$$

Since $t \le 1$, $C = [w,p] \subseteq [p-2, w+2] = I$, so $|C \cap I| = 2t$; and $A \cap C = [w,r]$, $C \cap D = [q,p]$ are subsets of C, of lengths $2(n_2 - s)$ and $2(s-n_1)$ with $n_1 = a + a'$, $n_2 = b + b'$, both nonnegative on the window. The bracket equals $2t - 2(n_2 - n_1) = 2t - 2(\delta_\mu + \delta_\nu) = -2\delta_\mu < 0$, and the first two terms are the two tents. □

Lemma 67 (Mixed offsets, pointwise)**.** *If $\delta_\mu \le 0 < \delta_\nu = t \le 1$, write $u = -\delta_\mu$, so $m_2 - m_1 = t + u$. Then for every s*

$$\ell(s) \;\le\; (2 - 2|s - m_1|)^+ + (2 - 2|s - m_2|)^+ + \min\bigl(2(s - m_1)^+,\; 2(m_2 - s)^+,\; 2\min(t, u)\bigr).$$

Proof. Outside the middle window the absorption of the previous proof applies verbatim (here $B = \emptyset$). On the window, $A \cap D = [q, r] = \emptyset$ since $\delta_\mu < 0$, and

$$\ell = |A \cap I| + |D \cap I| + c(s), \qquad c(s) = |C \cap I| - |A \cap C \cap I| - |C \cap D \cap I| = 2t - 2(n_2 - s)^+ - 2(s - n_1)^+,$$

by the same computations ($C \subseteq I$ from $t \le 1$; $A \cap C = [w, r]$ when $s \le n_2$ and empty otherwise, likewise $C \cap D$). Now $n_2 = m_1 + t$ and $n_1 = m_2 - t$ give $c \le 2t - 2(n_2 - s) = 2(s - m_1)$ and $c \le 2(m_2 - s)$; and $(n_2 - s)^+ + (s - n_1)^+ \ge (n_2 - n_1)^+ = (t - u)^+$ gives $c \le 2t - 2(t - u)^+ = 2\min(t, u)$. The first two terms are at most the two tents, as before. □

Integrating Lemma 67 over the band reduces everything to one two-variable inequality about the profile g.

Lemma 68 (The profile inequality)**.** *For $m_1 \le m_2$ set $D = m_2 - m_1$, $\tau^* = \min(1, D/2)$ and*

$$E(m_1, m_2) = \int_0^{\sqrt{2}} \min\bigl(2(s - m_1)^+,\; 2(m_2 - s)^+,\; 2\tau^*\bigr)\, ds.$$

Then $g(m_1) + g(m_2) + E(m_1, m_2) \le 2\bigl(2\sqrt{2} - 1\bigr)$, with equality exactly at $m_1 = m_2 = 1/\sqrt{2}$.

Proof. Four regions.

(i) $m_1, m_2 \in [\sqrt{2} - 1, 1]$. Here $g(m_i) = 2\sqrt{2} - 1 - 2x_i^2$ exactly, with $x_i = m_i - \frac{1}{\sqrt{2}}$, and $D \le 2(2 - \sqrt{2}) < 2$, so $\tau^* = D/2$ and E is at most the area of the unclipped profile, a triangle of base D and height D: $E \le D^2/2$. Since $D^2 = (x_2 - x_1)^2 \le 2(x_1^2 + x_2^2)$,

$$g(m_1) + g(m_2) + E \;\le\; 2(2\sqrt{2} - 1) - 2(x_1^2 + x_2^2) + (x_1^2 + x_2^2) \;\le\; 2(2\sqrt{2} - 1),$$

with equality only at $x_1 = x_2 = 0$.

(ii) $m_1 \le -1$. Then $g(m_1) = 0$, and the integrand of E is at most $\min(2(m_2 - s)^+, 2)$, so with $y = m_2$,

$$g(m_1) + g(m_2) + E \;\le\; h(y) := g(y) + \int_0^{\sqrt{2}} \min\bigl(2(y - s)^+, 2\bigr)\, ds.$$

Piecewise, from the explicit formulas for g: $h(y) = g(y) \le 2\sqrt{2} - 1$ for $y \le 0$; $h = 1 + 2y \le 2\sqrt{2} - 1$ on $[0, \sqrt{2} - 1]$; $h = 2\sqrt{2} - (y - \sqrt{2})^2 \le 2\sqrt{2}$ on $[\sqrt{2} - 1, \sqrt{2}]$; and $h = 2\sqrt{2}$ for $y \ge \sqrt{2}$. Since $2\sqrt{2} < 4\sqrt{2} - 2$, done.

(iii) $m_2 \ge \sqrt{2} + 1$. The substitution $s \mapsto \sqrt{2} - s$ carries $E(m_1, m_2)$ to $E(\sqrt{2} - m_2, \sqrt{2} - m_1)$ and g is symmetric, so this is case (ii).

(iv) The remainder $\{-1 \le m_1 \le m_2 \le \sqrt{2} + 1\}$ less the open square of (i). Certified: an adaptive covering in ball arithmetic verifies that the enclosure of $2(2\sqrt{2} - 1) - g(m_1) - g(m_2) - E$ is strictly positive on every box; 28 512 boxes suffice. The covering excludes only boxes inside $[0.4143, 1]^2 \subset (\sqrt{2} - 1, 1)^2$, a subset of region (i), so the four regions cover $\{m_1 \le m_2\}$. As a negative control, a box containing the equality point $(1/\sqrt{2}, 1/\sqrt{2})$ is refused, as it must be. □

Theorem 69 (The unconditional bound)**.** *Every ambidextrous moving sofa satisfies*

$$|S| \;\le\; 2\sqrt{2}-1 \;=\; 1.8284271\ldots$$

Proof. By (33) it suffices to bound the two-hallway area for every placement. If $\delta_\mu \le 0$ and $\delta_\nu \le 0$ this is Theorem 64. Otherwise, WLOG $t = \delta_\nu = \max > 0$. If $t \ge 1$, Lemma 65 gives $\sqrt{2} < 2\sqrt{2}-1$. If $t < 1$ and $\delta_\mu > 0$, Lemma 66 and Proposition 63 give $2\sqrt{2}-1$. If $t < 1$ and $\delta_\mu \le 0$, integrate Lemma 67 over the band: the tents give $\frac{1}{2}[g(m_1)+g(m_2)]$ and the flat-top term gives $\frac{1}{2}$ of an integral that is monotone in its height, so with $\min(t,u) \le \min(1, D/2) = \tau^*$ it is at most $\frac{1}{2}E(m_1,m_2)$; Lemma 68 finishes. □

Since Romik's sofa is ambidextrous, Theorem 69 brackets the ambidextrous supremum:

$$1.6449552\ldots \;=\; A_R^* \;\le\; A_{\text{ambi}} \;\le\; 2\sqrt{2}-1 = 1.8284271\ldots,$$

a factor of 1.1116. The previously available bound was 2.2195..., inherited from [1] because an ambidextrous sofa is in particular a one-corner sofa; we have found no bound specific to the ambidextrous problem in the literature.

The constant $2\sqrt{2}-1$ cannot be improved by adding angles to this pair: it is attained, by Proposition 61, so it is exactly the value of the two-hallway relaxation. The route to a smaller constant is more constraints, not a better estimate of these.

Four of the six constraints of Proposition 57 also force this value, and the surrounding structure is classical.

Proposition 70 (Four constraints suffice, and they recover Hammersley)**.** *Write L_θ, R_θ for a left- and a right-turning hallway at angle θ, with free placements. Then*

$$\sup\bigl|L_0 \cap L_{\pi/4} \cap L_{\pi/2}\bigr| = 2\sqrt{2}, \qquad \sup\bigl|L_0 \cap R_0 \cap L_{\pi/4} \cap R_{\pi/4}\bigr| \;\ge\; 2\sqrt{2}-1,$$

the first being Hammersley's classical bound for the one-corner problem.

The first equality is a regression check rather than a new result: it says the construction reproduces the known elementary bound $2\sqrt{2}$ for one corner, from three left-turning hallways. Adding the second handedness at $\pi/4$ to the corridor pair removes exactly 1. Since dropping constraints only enlarges the supremum, the four-constraint value already bounds the six-constraint one, so a proof of

$$\bigl|L_0 \cap R_0 \cap L_{\pi/4} \cap R_{\pi/4}\bigr| \;\le\; 2\sqrt{2}-1 \qquad \text{for all placements} \tag{37}$$

gives $A_{\text{ambi}} \le 2\sqrt{2}-1$ outright, with no branch and bound and no appeal to the remaining angles. That is what Theorem 64 and Lemmas 65–68 establish: (37) *holds*, since the corridor pair $L_0 \cap R_0$ is the strip whose rotated coordinates give the band, and the four sign cases of the offsets exhaust the placements. Theorem 69 is its restatement for sofas. Two structurally different placements attain the value: one splits the region into two congruent pieces of area $\sqrt{2}-\frac{1}{2}$, one leaves it connected. So $2\sqrt{2}-1$ *is* the supremum at those angles, attained by Proposition 61 and bounded above by Theorem 69, with no interval arithmetic in the two-hallway argument outside region (iv) of Lemma 68.

The construction was validated before use: intersecting along Σ's own corner path must give an area decreasing to $|\Sigma|$, and it does, through 2.050458, 2.000690, 1.991986, 1.990120, 1.645249 at $3, 9, 33, 129, 513$ angles per side for the two-family intersection. Two errors were caught by that check, one in the side of the inner corner quadrant and one in the identification of the mirrored family, before any bound was computed.

18 Relation to prior work

The one-corner problem was settled by Baek [1], who proved Gerver's sofa optimal. Our reading of that paper, for the purpose of separating what is his from what is not:

- The *arm lengths* f, g are his (Definition 6.2.1), defined through the contact points A and C; our α_1, α_2 are these shifted by the corridor width.
- His injectivity condition (Definition 6.1.2) has three parts. Part (1) requires the surface measure σ_K to be absolutely continuous on $[0, \pi/2)$ and on $(\pi/2, \pi]$, so an atom is permitted only at $\theta = \pi/2$; part (3) is $x_K'(t) \cdot u_t < 0$ and $x_K'(t) \cdot v_t > 0$ on $(0, \pi/2)$.
- Our (RC) shares part (1) verbatim and *adds* a bound on the density, while dropping part (3) entirely. No curvature bound appears in [1]; its only mentions of curvature are the standard identification of the surface-measure density with the radius of curvature, which we also use. The bound itself is classical, however: it is the sliding-ball condition of Remark 39, so (RC) is a known condition put to a new use rather than a new condition.
- His concavity argument attaches Mamikon regions to the overestimating region so that the total is linear in the cap; ours subtracts them from the cap. He works on a space of triples (K, B, D) of convex bodies; we work with the single support function H on $[0, \pi]$. Whether the two bookkeepings are equivalent we do not determine.
- The ambidextrous problem is not treated in [1], and we have found no upper bound for it in the literature.

The constants a_1, β, f_1 and the ODE families are Romik's [5]; the cubics of Section 7 are the standard ones and we claim no priority for them. This audit was made against [1] and the Kallus–Romik line [3] only, not against the wider literature.

19 The main theorem

Theorem 71. *Let $\mathcal{D}$ be the set of support functions H on $[0, \pi]$ of admissible caps such that*

(a) $H(0) = H(\pi/2) = 1$ (the gauge: horizontal translation, unit corridor)*;*

(b) (RC) *σ_K is absolutely continuous on $[0, \pi/2)$ and $(\pi/2, \pi]$ with density at most 1, the only atom being at $\theta = \pi/2$; equivalently, on $(0, \pi)$ $0 \le (H + H'')_{\mathrm{ac}} \le 1$ with no atom, together with $H'(0) = \frac{1}{2}$ and $H'(\pi) = -\frac{1}{2}$, the last two being the vanishing of the junction atoms of the ρ-extension (Lemma 85);*

(c) $c_y(t) = (H(t) - 1)\sin t + (H(t + \pi/2) - 1)\cos t \le \frac{1}{2}$ *for $t \in [0, \pi/2]$* (note $\le$, not $<$: this gives $|U \cap \rho U| = 0$, which is all an area bound needs)*;*

(d) $\alpha_1 < 0$ exactly on $[0, \beta)$ and $\alpha_2 > 0$ exactly on $[0, \pi/2 - \beta)$.

Then $\mathcal{D}$ is convex, $\Sigma \in \mathcal{D}$, and

$$|T| \;\le\; A_R^*$$

for every ambidextrous moving sofa T whose cap data lies in $\mathcal{D}$.

Proof. Each of (a)–(d) is a family of affine inequalities or equalities in H, by Lemma 18 and (17)–(19), so $\mathcal{D}$ is convex. Σ satisfies (b) by Proposition 51, (c) with margin $\frac{1}{2} - M = 0.1121\ldots$ by Theorem 8, and (d) by definition of β.

By (b) and Theorems 40 and 44 the combined sweep is injective, so $V = |N|$ by Proposition 35, whose proof also supplies the covering and containment that Proposition 20 assumes. By (c) and Lemma 1, $U \subseteq \{y \le \frac{1}{2}\}$ and $\rho U \subseteq \{y \ge \frac{1}{2}\}$, so $|U \cap \rho U| = 0$; with C_2 ρ-invariant this gives $|T| \le |C_2| - |N| - |\rho N| = |C_2| - 2|N| = Q$. (Corollary 2 states the strict form $M < \frac{1}{2}$, which is what Σ satisfies; the null-set form used here is what the hypothesis $\le$ delivers.) By (d) and Theorem 48, $\delta^2 Q \le 0$ on $\mathcal{D}$; by Theorem 45, $\delta Q(\Sigma) = 0$. A concave function with a critical point on a convex set attains its maximum there, so $Q(H) \le Q(\Sigma)$, and $Q(\Sigma) = A_R^*$ by Theorem 52. Combining, $|T| \le Q(H) \le A_R^*$. $\square$

20 The second variation

The bound above needs $\delta^2 Q \le 0$ on the cells the segment meets. This section establishes that, with the best constant, and on a family of cells rather than one.

Two routes to the constant are recorded and only the second is load-bearing. The first splits $[0, \pi/2]$ into $[0, \beta]$ and $[\beta, \pi/2]$, applies a separate Poincaré inequality on each, and discards the E_2 term, giving the weak constants of Remark 48. Treating each half as a *single* weighted eigenvalue problem removes both losses; not splitting at all is better still and gives the exact value.

Theorem 72 (Concavity, sharp order)**.** *On the cell $\mathcal{C}$, for every η with $\eta(0) = \eta(\pi/2) = 0$,*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -\tfrac{2}{3}\,\|\eta\|^2_{L^2(0,\pi)}. \tag{38}$$

Proof. Fix $\lambda \in (0,1)$ and $\kappa \in (0,1)$, and put $r = \lambda/(1-\lambda)$ and $q = 1 + 1/\kappa$. Two elementary inequalities dispose of the two coupling terms of (24). Since $(1+r)(1-\lambda) = 1$,

$$(1{+}r)a^2{+}2ab{+}(1{-}\lambda)b^2 = \left(\sqrt{1+r}\,a{+}\sqrt{1-\lambda}\,b\right)^2 \;\ge\; 0, \qquad \text{that is} \qquad -(a{+}b)^2 \;\le\; -\lambda\, b^2{+}r\, a^2,$$

applied with $a = \eta(t)$, $b = \eta'(t + \pi/2)$ on $E_2 = [0, \pi/2 - \beta)$; it buys gradient weight λ on $[\pi/2, \pi - \beta]$ at the price of mass r on $[0, \pi/2 - \beta)$. And $(x-y)^2 \le (1 + 1/\kappa)x^2 + (1+\kappa)y^2$ with $x = \eta(t + \pi/2)$, $y = \eta'(t)$ on $E_1 = [0, \beta)$ pays mass q on $[\pi/2, \pi/2 + \beta]$ and gradient weight $1 + \kappa$ on $[0, \beta)$. Substituting $s = t + \pi/2$ in both, the two halves decouple:

$$\tfrac{1}{2}\delta^2 Q[\eta] \;\le\; \int_0^{\pi/2} \left(m_L \eta^2 - w_L \eta'^2\right) + \int_{\pi/2}^{\pi} \left(m_R \eta^2 - w_R \eta'^2\right) \;=:\; B[\eta],$$

$$\begin{aligned} w_L &= 1 - \kappa \text{ on } [0, \beta),\ 2 \text{ after}, & m_L &= 2 + r \text{ on } [0, \tfrac{\pi}{2} - \beta),\ 2 \text{ after}, \\ w_R &= 1 + \lambda \text{ on } [\tfrac{\pi}{2}, \pi - \beta),\ 1 \text{ after}, & m_R &= 1 + q \text{ on } [\tfrac{\pi}{2}, \tfrac{\pi}{2} + \beta),\ 1 \text{ after}. \end{aligned}$$

Each half is a Rayleigh quotient, so $B[\eta] \le -\min(c_L, c_R)\,\|\eta\|^2_{L^2(0,\pi)}$, where c_L and c_R are the first eigenvalues of

$$-(w_L\eta')' {-} m_L \eta = c\,\eta \text{ on } (0, \tfrac{\pi}{2}), \quad \eta(0) = \eta(\tfrac{\pi}{2}) = 0; \qquad -(w_R\eta')' {-} m_R \eta = c\,\eta \text{ on } (\tfrac{\pi}{2}, \pi), \quad \eta(\tfrac{\pi}{2}) = 0,\ \eta'(\pi) = 0.$$

The Dirichlet conditions at 0 and $\pi/2$ are the gauge, not a restriction, and nothing pins $\eta(\pi)$.

Both coefficients are piecewise constant with three pieces, so the solution of the initial value problem $\eta = 0$, $w\eta' = 1$ at the left endpoint is a product of three transfer matrices acting on $(\eta,\, w\eta')^\top$,

$$T(\ell, w, m; c) = \begin{pmatrix} \cos k\ell & (wk)^{-1}\sin k\ell \\ -wk\sin k\ell & \cos k\ell \end{pmatrix}, \qquad k = \sqrt{(m+c)/w}$$

(hyperbolic when $m + c < 0$). Write $\Phi_L(c)$ for the resulting $\eta(\pi/2)$ and $\Phi_R(c)$ for $(w\eta')(\pi)$. Its zeros are exactly the eigenvalues, and by Sturm oscillation $\Phi > 0$ on $(-\infty, c_1)$, so a single sign evaluation bounds c_1 from *below*, which is the direction a certificate needs and the one a Rayleigh–Ritz or finite-element computation cannot supply.

Positivity alone does not locate c, because $\Phi > 0$ again on (c_2, c_3); one also needs $c < c_2$. That comes from min–max, crudely. Since $\int w\eta'^2 \geq w_{\min}\int \eta'^2$ and $-\int m\eta^2 \geq -m_{\max}\int \eta^2$, the form dominates the constant-coefficient one, so $c_k \geq w_{\min}\lambda_k - m_{\max}$ with λ_k the eigenvalues on an interval of length $\pi/2$, namely $(2k)^2$ for Dirichlet–Dirichlet and $(2k-1)^2$ for Dirichlet–Neumann. Hence

$$c_2^L \;\geq\; \tfrac{3}{4}\cdot 16 - \tfrac{63}{13} = 7.153\ldots, \qquad c_2^R \;\geq\; 1\cdot 9 - 6 = 3,$$

both above $\frac{2}{3}$. (At $k = 1$ the same bound gives -1.85 and -5, useless; the second eigenvalue is far enough away to be reached crudely while the first is not, which is exactly why the transfer matrix is needed.)

It remains to evaluate two signs. At $\kappa = \frac{1}{4}$, $\lambda = \frac{37}{50}$ and $c = \frac{2}{3}$ every weight and mass is rational ($w_L = \frac{3}{4}, 2, 2$ and $m_L = \frac{63}{13}, \frac{63}{13}, 2$; $w_R = \frac{87}{50}, \frac{87}{50}, 1$ and $m_R = 6, 1, 1$), so each $k^2 = (m+c)/w$ is exact, and the only inexact inputs are the three lengths β, $\frac{\pi}{2} - 2\beta$, β and the trigonometric functions of $k\ell$. In ball arithmetic at 300 bits, with $\beta = \arctan(u/2)$ enclosed from $u^3 + 3u = 2$ by exact rational bisection,

$$\Phi_L(\tfrac{2}{3}) \in 0.0046739486452026048\ldots \pm 6.4\cdot 10^{-73}, \qquad \Phi_R(\tfrac{2}{3}) \in 0.0026447861413929500\ldots \pm 8.3\cdot 10^{-73},$$

both certified strictly positive. Hence $c_L, c_R > \frac{2}{3}$, which is (38). □

Optimising the two parameters gives $\min(c_L, c_R) = 0.674944$ at $(\kappa, \lambda) = (0.26, 0.74)$. Since $\|\eta\|_{L^2(0,\pi/2)} \leq \|\eta\|_{L^2(0,\pi)}$, (38) also improves the $[0, \pi/2]$ constant of Theorem 48 from 0.371 to $\frac{2}{3}$.

The splitting into two scalar problems is itself avoidable, and avoiding it gives the sharp constant.

Theorem 73 (The sharp constant)**.** *The best constant in* (38) *is the first eigenvalue* c^* *of an explicit* 2×2 *Sturm–Liouville system, and*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\leq\; -\tfrac{73}{100}\,\|\eta\|^2_{L^2(0,\pi)}, \qquad c^* = 0.7309566\ldots \tag{39}$$

Proof. The two halves of $[0, \pi]$ are the same interval re-indexed. Put $p(t) = \eta(t)$ and $q(t) = \eta(t + \pi/2)$ for $t \in [0, \pi/2]$; then (24) reads $\frac{1}{2}\delta^2 Q = \int_0^{\pi/2} L$ with

$$L = 2p^2 - 2p'^2 + q^2 - q'^2 - \mathbf{1}_{E_2}(p + q')^2 + \mathbf{1}_{E_1}(q - p')^2,$$

and every boundary condition is forced: $p(0) = p(\pi/2) = 0$ is the gauge, $q(0) = \eta(\pi/2) = 0$ is the same gauge seen from the other half, and $q(\pi/2) = \eta(\pi)$ is free. Nothing has been discarded, so the first eigenvalue of this system is the best constant. Writing $M := -L$ as $ap'^2 + bq'^2 + gp^2 + dq^2 + 2e\,pq' + 2f\,qp'$,

piece	a	b	g	d	e	f
$[0, \beta)$	1	2	-1	-2	1	1
$[\beta, \frac{\pi}{2} - \beta)$	2	2	-1	-1	1	0
$[\frac{\pi}{2} - \beta, \frac{\pi}{2})$	2	1	-2	-1	0	0

On $[0, \beta)$, exactly where the obstruction sits, $e = f = 1$ and $2pq' + 2qp' = 2(pq)'$ is a total derivative: the E_1 obstruction and the E_2 resource cancel there up to the single boundary value $2p(\beta)q(\beta)$. The coupling is not pointwise, which is why every pointwise splitting loses and this one does not.

Eliminating p', q' in favour of the momenta $P = ap' + fq$ and $Q = bq' + ep$, the Euler–Lagrange system is first order in (p, P, q, Q) with piecewise constant matrix

$$A(c) = \begin{pmatrix} 0 & 1/a & -f/a & 0 \\ g - c - e^2/b & 0 & 0 & e/b \\ -e/b & 0 & 0 & 1/b \\ 0 & f/a & d - c - f^2/a & 0 \end{pmatrix},$$

so the transfer across a piece of length ℓ is $\exp\bigl(A(c)\ell\bigr)$. The left conditions $p(0) = q(0) = 0$ leave a two-dimensional solution space; propagating a basis through the three pieces and imposing $p(\pi/2) = 0$ and $Q(\pi/2) = 0$ gives a 2×2 determinant $\Phi(c)$ whose zeros are exactly the eigenvalues. A sign change brackets $c^* \in (0.7309, 0.7310)$.

For the certificate the oscillation argument is unavailable, since for a system the count is a Maslov index rather than a zero count, and it is not needed. Two facts compose: $c^* \geq \frac{2}{3}$ by Theorem 72, and Φ has no zero on $[\frac{2}{3}, \frac{73}{100}]$. The latter is checked by covering that interval with 60 balls and verifying in 400-bit ball arithmetic that no enclosure of Φ contains 0, with β enclosed from $u^3 + 3u = 2$ by exact rational bisection. Hence no eigenvalue lies at or below $\frac{73}{100}$. $\square$

The certified $\frac{73}{100}$ is 99.9% of c^*; the decoupled $\frac{2}{3}$ is 91.2%. Both are lower bounds, and no step in either is floating point. The covering route is also self-contained: with the crude spectral bound -4 of Theorem 76(c), covering $[-4, \frac{73}{100}]$ at the fixed lengths $(\beta, \frac{\pi}{2} - 2\beta, \beta)$ with 3074 boxes certifies $c_1 > \frac{73}{100}$ directly, without the decoupled chain or the oscillation argument; the negative control at $\frac{74}{100}$ is refused with the failing box at $c \in [0.7310, 0.7313]$, which lies just above $c^* = 0.7309566\ldots$ and within $4 \cdot 10^{-5}$ of it: the enclosure of Φ there is still wide enough to contain 0, which is how a target past c^* fails. The decoupled chain remains the hand-checkable route to $\frac{2}{3}$. Nothing in the argument stops at $\frac{73}{100}$: certifying a target c needs the enclosures to separate from 0 on every subinterval of $[\frac{2}{3}, c]$, and those must shrink as c approaches the root, so the number of balls grows: $60, 60, 800, 12000$ for targets $0.73, 0.7305, 0.7309, 0.73095$. The last is 99.9991% of c^*. We quote $\frac{73}{100}$ because a reader can check it with 60 balls; the limit is arithmetic cost, not a gap.

Remark 74 (The $[\pi/2, \pi]$ eigenvalue in closed form)**.** When only the weight is raised and the mass is left at 1 – that is, $w = 1 + \lambda_J$ on $[\pi/2, \pi - \beta]$ and 1 on $[\pi - \beta, \pi]$ – the transfer product collapses to a transcendental equation for the first eigenvalue $\Lambda(\lambda_J)$ of $-(w\eta')' = \Lambda\eta$ with $\eta(\pi/2) = 0$, $\eta'(\pi) = 0$:

$$\sqrt{w_1}\,\cot\bigl(L_1\sqrt{\Lambda/w_1}\bigr) = \tan\bigl(L_2\sqrt{\Lambda}\bigr), \qquad w_1 = 1 + \lambda_J, \quad L_1 = \tfrac{\pi}{2} - \beta, \quad L_2 = \beta. \tag{40}$$

At $\lambda_J = 0$ this reads $\cot(L_1\sqrt{\Lambda}) = \cot(\pi/2 - L_2\sqrt{\Lambda})$, so $\sqrt{\Lambda}\,(L_1 + L_2) = \pi/2$; since $L_1 + L_2 = \pi/2$ *exactly*, $\Lambda(0) = 1$ exactly. That is the marginal case $P_3 = 1$ of Theorem 48, and it explains why no gain is available on $[\pi/2, \pi]$ until the weight is raised. For $\lambda_J > 0$ put $G(w_1) = \sqrt{w_1}\cot(L_1/\sqrt{w_1}) - \tan\beta$. Then $G(1) = 0$, and G increases: the prefactor increases, while $L_1/\sqrt{w_1}$ decreases and cot decreases on $(0, \pi)$. So $G > 0$ for $w_1 > 1$. The left side of (40) decreases in Λ and the right side increases, so the root moves right and $\Lambda(\lambda_J) > 1$ for every $\lambda_J > 0$:

λ_J	0	0.05	0.10	0.16483	0.25	0.35
Λ	1	1.049467	1.098883	1.162878	1.246819	1.345183

Theorem 72 runs this mechanism on both halves at once, raising the mass as well as the weight.

Concavity on every ordered anchored cell

The concavity results so far concern single cells: Σ's own (Theorems 48, 72, 73) and the eleven measured entries of Proposition 91 below. This subsection proves concavity for the whole ordered anchored family at once. In the variables $p(t) = \eta(t)$, $q(t) = \eta(t + \pi/2)$ on $[0, \pi/2]$, with $p(0) = p(\pi/2) = q(0) = 0$ and $q(\pi/2)$ free, the cell form is

$$B_{E_1,E_2}[\eta] = \int_0^{\pi/2} \big[2p^2 - 2p'^2 + q^2 - q'^2\big] - \int_{E_2} (p + q')^2 + \int_{E_1} (q - p')^2, \tag{41}$$

where $E_1 = \{\alpha_1 < 0\}$ and $E_2 = \{\alpha_2 > 0\}$.

Lemma 75 (Monotonicity in the sign sets)**.** *If $E_1 \subseteq E_2$ then $B_{E_1,E_2}[\eta] = B_{E_2,E_2}[\eta] - \int_{E_2 \setminus E_1} (q - p')^2 \le B_{E_2,E_2}[\eta]$.*

Proof. The two forms differ only in the set carrying the $(q - p')^2$ term. □

So every cell with $E_1 \subseteq E_2$ and E_2 *anchored* is dominated by a member of the one-parameter diagonal family $D_\tau := B_{[0,\tau),[0,\tau)}$, and it suffices to treat that family.

Theorem 76 (Diagonal concavity)**.** *$D_\tau[\eta] \le 0$ for every $\tau \in [0, \pi/2]$ and every admissible η.*

Proof. On $[0, \tau)$ the integrand of (41) combines, by the algebra of Theorem 73, into $(p^2 - p'^2) + 2(q^2 - q'^2) - 2(pq)'$, so with $p(0)q(0) = 0$,

$$D_\tau = \int_0^\tau \big[(p^2 - p'^2) + 2(q^2 - q'^2)\big] - 2p(\tau)q(\tau) + \int_\tau^{\pi/2} \big[2(p^2 - p'^2) + (q^2 - q'^2)\big]. \tag{42}$$

Three overlapping ranges of τ are treated by three arguments.

Range (a): $\tau \le 1/\sqrt{3}$. Write (42) as $B_{\text{bare}} + \int_0^\tau [(p'^2 - p^2) + (q^2 - q'^2)] - 2p(\tau)q(\tau)$ with $B_{\text{bare}} := \int_0^{\pi/2} [2(p^2 - p'^2) + (q^2 - q'^2)]$. The Poincaré constants for the boundary conditions are 4 (Dirichlet–Dirichlet, for p) and 1 (Dirichlet–Neumann, for q), so $B_{\text{bare}} \le -\frac{3}{2}\int_0^{\pi/2} p'^2$. The added terms are bounded by $\int_0^\tau p'^2 \le \int_0^{\pi/2} p'^2$, by $\int_0^\tau q^2 \le \tau^2 \int_0^\tau q'^2$ (from $q(t)^2 \le t\int_0^t q'^2$), and, using Young's inequality at weight $s = 1/(2\tau)$ together with $p(\tau)^2 \le \tau\int_0^\tau p'^2$ and $q(\tau)^2 \le \tau\int_0^\tau q'^2$,

$$2|p(\tau)q(\tau)| \;\le\; \frac{p(\tau)^2}{2\tau} + 2\tau\, q(\tau)^2 \;\le\; \tfrac{1}{2}\int_0^{\pi/2} p'^2 + 2\tau^2 \int_0^\tau q'^2.$$

Collecting, $D_\tau \le \left(-\frac{3}{2}+1+\frac{1}{2}\right)\int_0^{\pi/2} p'^2 + (3\tau^2-1)\int_0^\tau q'^2 \le 0$ for $\tau^2 \le \frac{1}{3}$.

Range (b): $\sigma := \pi/2-\tau \le \frac{1}{18}$. The obstruction here is that the bound must degenerate: $D_{\pi/2}$ annihilates the mode $(p,q)=(0,\sin t)$, so no estimate with a uniform negative constant exists near $\tau=\pi/2$, and the argument must see the mode. Expand $q=a\varphi_1+r$ in the Dirichlet–Neumann eigenbasis $\varphi_k=\sin((2k-1)t)$ of $[0,\pi/2]$, with $r\perp\varphi_1$ in both inner products, and set $D := \int q'^2 - \int q^2 \ge 0$. From the eigenvalues $(2k-1)^2$,

$$\int r'^2 \le \tfrac{9}{8}D, \qquad \|r\|_{L^2}^2 \le \tfrac{D}{8}, \qquad r(t)^2 \le t\int_0^t r'^2 \le \tfrac{9\pi}{16}D. \tag{43}$$

Since $\varphi_1''=-\varphi_1$ and $r(0)=0$, integration by parts gives $\int_0^\tau(\varphi_1 r-\varphi_1' r') = -\varphi_1'(\tau)r(\tau) = -\sin\sigma\, r(\tau)$, and $\int_0^\tau(\varphi_1^2-\varphi_1'^2) = -\frac{1}{2}\sin 2\sigma$. Hence, using (43) and Young's inequality $2XY \le 3X^2+\frac{1}{3}Y^2$ on the middle term,

$$2\int_0^\tau(q^2-q'^2)+\int_\tau^{\pi/2}(q^2-q'^2) \;\le\; -\Big(\tfrac{7}{8}-\tfrac{3\pi}{16}\Big)D - a^2\sin\sigma(\cos\sigma-3\sin\sigma) \;\le\; -\tfrac{2}{7}D-\tfrac{4}{5}\sigma a^2,$$

the last step using $\frac{7}{8}-\frac{3\pi}{16}\ge\frac{2}{7}$ and $\sin\sigma(\cos\sigma-3\sin\sigma)\ge\sigma(1-3\sigma-\sigma^2)\ge\frac{4}{5}\sigma$ for $\sigma\le\frac{1}{18}$. For p, with $P^2 := \int_\tau^{\pi/2}p'^2$ and the Dirichlet Poincaré constant $(2\sigma/\pi)^2 \le \frac{1}{400}$ on $[\tau,\pi/2]$,

$$\int_0^\tau(p^2-p'^2)+2\int_\tau^{\pi/2}(p^2-p'^2) \;\le\; -\tfrac{3}{4}\int_0^{\pi/2}p'^2 - \left(1-\tfrac{1}{400}\right)P^2 \;\le\; -\tfrac{174}{100}P^2.$$

The boundary term couples the pieces through $p(\tau)^2\le\sigma P^2$ and $q(\tau)=a\cos\sigma+r(\tau)$:

$$2|p(\tau)||a|\cos\sigma \le \tfrac{5}{4}P^2+\tfrac{4}{5}\sigma\cos^2\!\sigma\, a^2, \qquad 2|p(\tau)||r(\tau)| \le \tfrac{9}{25}P^2+\tfrac{25}{9}\,\sigma\cdot\tfrac{9\pi}{16}D,$$

and $\frac{25}{9}\cdot\frac{9\pi}{16}\,\sigma = \frac{25\pi}{16}\sigma \le \frac{25\pi}{288} \le \frac{2}{7}$ for $\sigma\le\frac{1}{18}$. Summing the three displays, the coefficients of a^2, D and P^2 are respectively $\frac{4}{5}\sigma(\cos^2\sigma-1)\le 0$, at most $-\frac{2}{7}+\frac{25\pi}{288}\le 0$, and $-\frac{174}{100}+\frac{5}{4}+\frac{9}{25}=-\frac{13}{100}<0$. Hence $D_\tau\le 0$.

Range (c): $\tau\in[0.55,\,1.5153]$. The form D_τ is the 2×2 Sturm–Liouville system of Theorem 73 with pieces of lengths τ, $\pi/2-\tau$ and coefficient rows $(1,2,-1,-2,1,1)$, $(2,1,-2,-1,0,0)$, so its eigenvalues are the zeros of the same shooting determinant $\Phi(c;\tau)$. Two facts combine. First, the spectrum is bounded below by -4: pointwise, $2pq' \ge -(2p^2+\frac{1}{2}q'^2)$ and $2qp' \ge -(2q^2+\frac{1}{2}p'^2)$ absorb the cross terms of $-D_\tau$ leaving non-negative derivative coefficients, so $-D_\tau[\eta] \ge -\int(3p^2+4q^2) \ge -4\|\eta\|^2$. Second, $\Phi(c;\tau)\ne0$ for every $(c,\tau)\in[-4,0]\times[0.55,1.5153]$: the rectangle is covered adaptively by 305 boxes on which the ball-arithmetic enclosure of Φ excludes zero, with the lengths carried as balls containing $[\tau_{\mathrm{lo}},\tau_{\mathrm{hi}}]$. Hence no eigenvalue lies in $[-4,0]$ and none lies below -4, so the first eigenvalue is positive and $D_\tau\le0$ (indeed <0) on this range.

The ranges $[0,1/\sqrt{3}]$, $[0.55,1.5153]$ and $[\pi/2-\frac{1}{18},\pi/2]$ overlap and cover $[0,\pi/2]$. ☐

The covering was validated by negative controls: enlarged c-ranges crossing the measured first eigenvalue are refused, with the failing boxes localising the eigenvalue curve to $(\tau,c)=(0.785,0.377)$ for the mirror family below and $(1.514,0.041)$ for the diagonal, matching the measured values.

Corollary 77 (Ordered anchored concavity)**.** *$\delta^2Q\le0$ on every cell whose sign sets satisfy $\{\alpha_1<0\}\subseteq\{\alpha_2>0\}$ with $\{\alpha_2>0\}=[0,\tau)$ anchored. The set $\{\alpha_1<0\}$ need not be an interval.*

Theorem 78 (Uniform constant on the mirror family)**.** *For every $\tau \in [0, \pi/4]$ the mirror-anchored cell $(\tau, \pi/2 - \tau)$ satisfies*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -\tfrac{3}{10}\,\|\eta\|^2_{L^2(0,\pi)}.$$

Proof. The cell form is the system of Theorem 73 with lengths τ, $\pi/2 - 2\tau$, τ; Σ's cell is $\tau = \beta$. The lower bound -4 on the spectrum holds as in range (c), and the covering of $[-4, \frac{3}{10}] \times [0, 0.7854]$ by 249 boxes shows $\Phi \neq 0$ there; since $0.7854 > \pi/4$, every $\tau \in [0, \pi/4]$ is included. (For τ slightly past $\pi/4$ the middle length crosses zero and the transfer formula is an analytic continuation of no further relevance; the enclosure covers every real τ in each ball, in particular the range where the formula is the shooting determinant of the cell.) Hence the first eigenvalue exceeds $\frac{3}{10}$ on the whole family. Measured, it decreases from 0.875 at $\tau = 0$ to 0.370 at $\tau = \pi/4$, with 0.7331 at Σ's cell. ☐

21 Stability and uniqueness

Because Q is exactly quadratic on the cell and $\delta Q(\Sigma) = 0$, the same ingredients give more than a bound.

Theorem 79 (Stability)**.** *For every ambidextrous moving sofa T whose cap data $H = H_\Sigma + \eta$ lies in $\mathcal{D}$,*

$$|T| \;\le\; A_R^* - \tfrac{73}{100}\,\|\eta\|^2_{L^2(0,\pi)}. \tag{44}$$

Proof. On the cell $\mathcal{C}$ the functional Q is a quadratic polynomial in H, so $Q(H_\Sigma + \eta) = Q(\Sigma) + \delta Q(\Sigma)[\eta] + \frac{1}{2}\delta^2 Q[\eta]$ exactly. The first variation vanishes by Theorem 45 and $Q(\Sigma) = A_R^*$ by Theorem 52, while Theorem 73 bounds the second variation. Combining with $|T| \le Q(H)$ from Theorem 71 gives (44). ☐

Corollary 80 (Uniqueness)**.** *Σ is the unique ambidextrous moving sofa of area A_R^* whose cap data lies in $\mathcal{D}$; every other one has strictly smaller area, by the amount (44).*

Proof. Equality in (44) forces $\|\eta\|_{L^2(0,\pi/2)} = 0$, and then $\eta \equiv 0$ on $[0, \pi/2]$; Theorem 48 shows $\delta^2 Q < 0$ strictly unless $\eta \equiv 0$ throughout, since with $\eta \equiv 0$ on $[0, \pi/2]$ the E_2 term of (24) contributes $-\int_0^{\pi/2-\beta} \eta'(t + \pi/2)^2 dt < 0$ unless η is constant on $[\pi/2, \pi]$, and $\eta(\pi/2) = 0$ makes that constant zero. ☐

The domain $\mathcal{D}$ is cut out in part by (RC), and (RC) enters at exactly one point: it forces the flux excess $E := V - |N|$ to vanish, so that Q is tight. Separation and Reynolds give, with no curvature hypothesis,

$$|T| \;=\; |C_2| - 2|N| \;=\; Q + 2E, \qquad E \ge 0. \tag{45}$$

It is natural to ask for a concave majorant of E, which would replace $\{E = 0\}$ by something larger. There is none.

Proposition 81 (No concave majorant)**.** *Let $\mathcal{D}'$ be convex with Σ in its algebraic interior, and let $\widetilde{E} : \mathcal{D}' \to \mathbb{R}$ be concave with $\widetilde{E} \ge 0$ and $\widetilde{E}(\Sigma) = 0$. Then $\widetilde{E} \equiv 0$, and hence $E \equiv 0$ on $\mathcal{D}'$.*

Proof. Let $a, b \in \mathcal{D}'$ with $\Sigma = \frac{1}{2}(a + b)$. Concavity gives $0 = \widetilde{E}(\Sigma) \ge \frac{1}{2}\widetilde{E}(a) + \frac{1}{2}\widetilde{E}(b) \ge 0$, so $\widetilde{E}(a) = \widetilde{E}(b) = 0$. As Σ is interior, every point of $\mathcal{D}'$ arises this way. ☐

So any architecture with a concave, tight majorant is confined to the injectivity locus. The hypothesis to weaken is concavity of $\widetilde{E}$: what the argument needs is that $Q + 2\widetilde{E}$ be concave, and the strict concavity of Q leaves room.

Theorem 82 (Excess budget)**.** *Let $\mathcal{D}'$ be convex with $\Sigma \in \mathcal{D}'$ and $\theta \in [0,1]$. Suppose $M < \frac{1}{2}$ on $\mathcal{D}'$, that $\frac{1}{2}\delta^2 Q \leq -c\|\eta\|^2_{L^2(0,\pi)}$ there, and that*

$$E(H) \;\leq\; \tfrac{1}{2}\,\theta\, c\, \|H - H_\Sigma\|^2_{L^2(0,\pi)} \qquad \text{on } \mathcal{D}'. \tag{46}$$

Then $|T| \leq A^*_R - (1-\theta)\, c\, \|H - H_\Sigma\|^2_{L^2(0,\pi)}$ for every ambidextrous moving sofa T with cap data in $\mathcal{D}'$.

Proof. Concavity of Q with $\delta Q(\Sigma) = 0$ gives $Q(H) \leq A^*_R - c\|\eta\|^2$. Adding (45) and (46), $|T| = Q + 2E \leq A^*_R - c\|\eta\|^2 + \theta c\|\eta\|^2$. □

At $\theta = 0$ this is Theorem 79. For $\theta > 0$ it admits $E > 0$, so Proposition 81 is evaded, and the room available is proportional to c: the constant $c^* = 0.7309566$ of Theorem 73 is what makes (46) usable. Measured along $H_\Sigma + \varepsilon\phi$ with ϕ the bump at $\theta_0 = 1$ of half-width 0.45, the least admissible θ is

ε	0.0141	0.0200	0.0283	0.0400	0.0566	0.0800
θ	0.100	0.301	0.683	1.276	2.090	3.106

so $\theta < 1$ holds out to $\varepsilon \approx 0.031$, against $\varepsilon_c = 0.006492$ for (RC): the admissible radius grows by a factor of about 4.8 in that direction. This is a measurement, not a proof.

Theorem 83 (Stability on the mirror class)**.** *Let T be a connected ambidextrous moving sofa whose cap data H satisfies the gauge, the mirror condition (48), and (RC), and suppose that along the segment $H_s = (1-s)H_\Sigma + sH$ the set $\{\alpha_1(\cdot; H_s) < 0\}$ is an interval $[0, \tau_1(s))$ with $\tau_1(s) \leq \pi/4$ for every s. Then*

$$|T| \;\leq\; A^*_R - \tfrac{3}{10}\, \|H - H_\Sigma\|^2_{L^2(0,\pi)},$$

and equality forces $H = H_\Sigma$.

Proof. The mirror condition is linear and holds at H_Σ, so every H_s satisfies it, and by Lemma 89 each cell met by the segment is mirror-anchored with parameter $\tau_1(s) \leq \pi/4$. By Theorem 78, $g(s) := Q(H_s)$ has $g''(s) \leq -\frac{3}{5}\|H - H_\Sigma\|^2$ on each cell, g is C^1 across cell boundaries as in Proposition 91, $g(0) = A^*_R$ and $g'(0) = 0$; integrating, $Q(H) \leq A^*_R - \frac{3}{10}\|H - H_\Sigma\|^2$. Connectedness gives $M \leq \frac{1}{2}$ (Theorem 11), and (RC) gives $V = |N|$, so $|T| = |C_2| - 2|N| = Q(H)$. □

Lemma 84 (The arm sandwich)**.** *Suppose $0 \leq (H + H'')_{\mathrm{ac}} \leq 1$ on $(0,\pi)$, with atoms only at $\pi/2$. Then on $(0, \pi/2)$, in the absolutely continuous sense,*

$$\alpha_2 \;\leq\; \alpha_1' \;\leq\; \alpha_2 + 1, \qquad -\alpha_1 - 1 \;\leq\; \alpha_2' \;\leq\; -\alpha_1.$$

The upper bounds use convexity ($H + H'' \geq 0$), the lower bounds use (RC).

Proof. $\alpha_1' = G' - F''$ and $-F'' \in [F - 1,\, F]$ by the two-sided bound on $F + F''$, so $\alpha_1' \in [G' + F - 1,\, G' + F] = [\alpha_2,\, \alpha_2 + 1]$. Likewise $\alpha_2' = F' + G''$ with $G'' \in [-G,\, 1 - G]$ gives $\alpha_2' \in [-\alpha_1 - 1,\, -\alpha_1]$. □

The pair (α_1, α_2) therefore rotates: where $\alpha_2 > 0$ the first arm strictly increases, and where $\alpha_1 > 0$ the second strictly decreases. This is the differential form of the oscillator mechanism behind Theorem 40, and it makes the sign hypothesis of Theorem 83 checkable from the arms alone. It does not make it removable:

Lemma 85 (Forced boundary derivatives)**.** *Under (RC), $H'(0) = \frac{1}{2}$ and $H'(\pi) = -\frac{1}{2}$.*

Proof. C_2 is ρ-symmetric, so its support function satisfies $h(\theta) = h(-\theta) + \sin\theta$ on the whole circle; this determines h on $[\pi, 2\pi]$ from H on $[0, \pi]$. On $(\pi, 2\pi)$, writing $\psi = 2\pi - \theta$, one finds $h + h'' = (H + H'')(\psi)$, so no condition arises there. At the two junctions the extension has derivative jumps

$$h'(0^+) - h'(0^-) = 2H'(0) - 1, \qquad h'(\pi^+) - h'(\pi^-) = -2H'(\pi) - 1,$$

which are the atoms of the surface area measure at $\theta = 0$ and $\theta = \pi$. Convexity forces both to be non-negative; (RC) permits atoms only at $\pm\pi/2$, so both vanish. $\square$

For Σ these read 0.4999999990 and -0.4999999987 in the computed data, the deviation being the finite-difference error. Lemma 85 is what makes the class of admissible caps rigid at the endpoints, and it is what selects a single member of the following family.

Remark 86 (The sign hypothesis is not removable)**.** Consider $H_c(\theta) = 1 - c + c(\sin\theta + \cos\theta)$, the support data of the disc of radius $1 - c$ centred at (c, c). It satisfies the gauge and the mirror condition (48) with $A = 2c$, and $H_c + H_c'' \equiv 1 - c$, so the absolutely continuous part of (RC) holds for every $c \in (0, 1)$. By Lemma 85 only $c = \frac{1}{2}$ is admissible: $H_c'(0) = c$, and $c < \frac{1}{2}$ makes the junction atoms negative, so the ρ-extension is not convex at all, while $c > \frac{1}{2}$ creates atoms at 0 and π that (RC) forbids. At $c = \frac{1}{2}$ the body is the disc of radius $\frac{1}{2}$ centred at $(\frac{1}{2}, \frac{1}{2})$, the incircle of the unit square, and every condition holds with $H + H'' \equiv \frac{1}{2}$. Its arms are

$$\alpha_1 \equiv \alpha_2 \equiv -\tfrac{1}{2},$$

so $\{\alpha_1 < 0\} = [0, \pi/2)$ while $\{\alpha_2 > 0\} = \emptyset$, and the ordering fails identically. Moreover $|N| = 0$ and $|C_2| = \pi/4$ (both the cap formula and a direct polygon computation give 0.785398163), so the maximal ambidextrous sofa with this cap datum is the disc itself, and a disc of diameter 1 moves freely through the corridor and turns either corner. The datum is therefore realized by a connected ambidextrous moving sofa. Hence the ordered anchored sign structure follows neither from the gauge, mirror, convexity and (RC), nor from realizability, and the hypothesis of Theorem 83 is permanent rather than provisional.

Remark 87 (A hypothesis of the niche formula, made explicit)**.** The disc also isolates an assumption implicit in (20). The face-1 segment is $s \in [\alpha_1^+, \sigma]$, so its contribution is $\frac{1}{2}\big((\sigma - \alpha_1^+)^+\big)^2$, and writing it as $\frac{1}{2}(\sigma - \alpha_1)^2$ presupposes

$$\sigma \;\geq\; \alpha_1^+ \qquad \text{on } [0, \pi/2]. \tag{47}$$

For the disc $\alpha_1^+ = 0$ while σ decreases to $-\frac{1}{2}$, and the unmodified expression returns $V = -0.1427$ against $|N| = 0$, in apparent conflict with Proposition 28; with the segment truncated, $V = 0 = |N|$ and the conflict disappears. For Σ, (47) holds: $\sigma - \alpha_1 = \cos t\, x'(t)$ by Proposition 49 and x is strictly increasing by Proposition 30, while $\sigma \geq 0$ throughout; the computed minimum of $\sigma - \alpha_1^+$ is $+2.5 \cdot 10^{-7}$, attained as $t \to \pi/2$. All uses of (20) in this note are on data satisfying (47).

Proposition 88 (An explicit ball inside the domain)**.** *Let $r_0 = \frac{1}{20}$. Every η with $\eta(0) = \eta(\pi/2) = 0$, satisfying the mirror condition, and with $\max(\|\eta\|_\infty, \|\eta'\|_\infty, \|\eta''\|_\infty) \le r_0$, gives $H = H_\Sigma + \eta$ satisfying (RC), $M(H) < \frac{1}{2}$, and the sign hypothesis of Theorem 83 along the entire segment $[H_\Sigma, H]$. In particular the stability bound applies to every connected ambidextrous sofa with such cap data.*

Proof. (RC): $\max(H_\Sigma + H''_\Sigma)_{\mathrm{ac}} = \frac{3}{4} f_1 \sqrt{4 - 2\sqrt{2}} \cos(\frac{\beta}{2} + \frac{\pi}{8}) = 0.8389\ldots$ in closed form, and $\|\eta + \eta''\|_\infty \le 2r_0 = 0.1 \le 0.1611$. Ceiling: $M(\Sigma) = 0.38784\ldots$ in closed form, and $|\delta c_y| \le \|\eta\|_\infty(\sin t + \cos t) \le \sqrt{2}\, r_0 = 0.0708$, so $M(H) \le 0.4586 < \frac{1}{2}$. Signs: the perturbations obey $|\delta\alpha_i| \le \|\eta\|_\infty + \|\eta'\|_\infty \le 2r_0 = 0.1$, and six margins of the unperturbed arms are certified on grids of spacing $h = 5 \cdot 10^{-4}$ with the Lipschitz constant $L = 2$ supplied by Lemma 84: writing $S := \sup(|\alpha_1| \vee |\alpha_2|)$, the sandwich gives $\sup|\alpha'_i| \le S + 1$, and the grid maximum 0.7506 bootstraps to $S \le 0.7506 + (S+1)h/2$, hence $S \le 0.7513$ and $L \le 1.76 \le 2$. The certified margins are: $\alpha_1 \le -g$ on $[10^{-3}, \beta - d]$ and $\alpha_1 \ge g$ on $[\beta + d, \pi/2]$; $\alpha_2 \ge g$ on $[10^{-3}, \pi/2 - \beta - d]$ and $\alpha_2 \le -g$ on $[\pi/2 - \beta + d, \pi/2]$; and $\alpha_2 \ge 0.5975$, $\alpha_1 \ge 0.5975$ on the windows $[\beta - d, \beta + d]$, $[\pi/2 - \beta - d, \pi/2 - \beta + d]$ respectively, with $d = 0.15$ and $g = 0.1224 > 2r_0$. Outside the windows the perturbed arms keep their signs. Inside α_1's window, Lemma 84 gives $\alpha'_1 \ge \alpha_2 \ge 0.5975 - 0.1 > 0$ for the perturbed data, so α_1 crosses zero exactly once there; likewise $\alpha'_2 \le -\alpha_1 \le -(0.5975 - 0.1) < 0$ in its window. So $\{\alpha_1 < 0\} = [0, \tau_1)$ with $\tau_1 < \beta + d \le \pi/4$ and $\{\alpha_2 > 0\} = [0, \tau_2)$ with $\tau_2 > \pi/2 - \beta - d > \beta + d$, ordered and anchored. Replacing η by $s\eta$, $s \in [0,1]$, only shrinks the perturbation, so the same holds along the segment. $\square$

The grid evaluations use double precision; the margins exceed the evaluation error by eleven orders of magnitude, and the Lipschitz constant is proved, not sampled. This replaces, in the load-bearing direction, the numerical statement that $\mathcal{D}$ is a C^2-neighbourhood: the domain *contains* the explicit C^2-ball of radius $\frac{1}{20}$ within the mirror class. The outer containment (that the radius along $\sin(2k\theta)$ shrinks like k^{-2}) remains a measurement.

The hypotheses of Theorem 83 are non-perturbative: no smallness of $H - H_\Sigma$ is assumed, and the class contains data as far from Σ as the mirror caps with τ_1 anywhere in $[0, \pi/4]$. The binding smallness constraint that remains in $\mathcal{D}$ is (RC) alone.

The sign cell is the least structural part of $\mathcal{D}$. On a natural linear subclass it collapses to a single scalar inequality.

Lemma 89 (Mirror caps)**.** *Suppose the cap is symmetric about some vertical line, equivalently*

$$H(\theta) - H(\pi - \theta) = A\cos\theta, \qquad A := H(0) - H(\pi). \tag{48}$$

Then $\alpha_2(\pi/2 - t) = \alpha_1(t)$ for every $t \in [0, \pi/2]$. Consequently $E_2 = \frac{\pi}{2} - E_1$ as sets, and if $E_1 = [0, \tau_1)$ then $E_2 = [0, \frac{\pi}{2} - \tau_1)$, so

$$\tau_1 + \tau_2 = \tfrac{\pi}{2}.$$

Proof. Condition (48) says $h(u) - h(\sigma u) = \langle (A, 0), u\rangle$ for the reflection $\sigma(x, y) = (-x, y)$, that is, the cap coincides with its mirror image translated by $(A, 0)$; the axis is $x = A/2$. Replacing θ by $\pi/2 + t$ in (48) gives $H(\pi/2 - t) = H(\pi/2 + t) + A\sin t$, and differentiating (48) gives $H'(\pi - \theta) = -H'(\theta) - A\sin\theta$. Hence

$$\alpha_2(\tfrac{\pi}{2} - t) = H(\tfrac{\pi}{2} - t) - 1 + H'(\pi - t) = \big[H(\tfrac{\pi}{2} + t) + A\sin t\big] - 1 + \big[-H'(t) - A\sin t\big] = H(\tfrac{\pi}{2} + t) - 1 - H'(t) = \alpha_1(t).$$

$\square$

Condition (48) is linear in H once A is treated as a free parameter, so the mirror caps form a linear subspace and intersecting $\mathcal{D}$ with it preserves convexity. Both Σ and Gerver's cap satisfy it, to $7 \cdot 10^{-16}$ in the computed data. On that subclass the anchored sign pattern is fixed by the single number τ_1, and the order condition $\tau_1 \leq \tau_2$ of Proposition 91 becomes

$$\tau_1 \;\leq\; \tfrac{\pi}{4},$$

which at Σ reads $\beta = 0.289654 \leq 0.785398$, a margin of 0.4957. Lemma 89 is also what makes $L_1 + L_2 = \pi/2$ exact in (40), hence $\Lambda(0) = 1$ exactly: the two lengths there are τ_2 and τ_1.

Remark 90 (Why Gerver's cap is excluded)**.** Gerver's cap satisfies (48) and, by Theorem 82, the excess budget along the whole segment $[H_\Sigma, H_\mathrm{G}]$ at $\theta = 0.0072$; and (RC) fails on that segment only at $s = 0.398$. None of these is what excludes it. Along the segment $M = \max_t c_y(t)$ crosses $\frac{1}{2}$ at $s = 0.406$, and at Gerver $M = 0.6643$, so by Theorem 11 any ambidextrous sofa with that corner path omits a vertical strip of width 0.1646 and a connected one lies entirely on one side of it, contradicting that it meets both arms. So *no connected ambidextrous moving sofa has Gerver's cap data*: the exclusion is a property of the competitor class, not a limitation of the curvature hypothesis. Weakening (RC) cannot reach Gerver, and should not.

Proposition 91 (Beyond Σ's own cell)**.** *Q is C^1 and piecewise quadratic, so if every cell met by the segment $[H_\Sigma, H]$ has $\delta^2 Q \leq 0$, then Q is concave along that segment and $Q(H) \leq Q(\Sigma) + \delta Q(\Sigma)[H - H_\Sigma] = A^*_R$. Convexity of the union of those cells is not needed. In particular this holds whenever the segment meets only cells with $\{\alpha_1 < 0\} \subseteq \{\alpha_2 > 0\}$ and $\{\alpha_2 > 0\}$ anchored, which is no longer a measured condition: Corollary 77 proves concavity on every such cell. The eleven sample values $-0.8751, -0.1404, -0.3602, -0.5397, -0.4497, -0.3431, -0.2285, -0.1074, -0.0457, -0.7876, -0.042$ at $(\tau_1, \tau_2) = (0, \frac{\pi}{2}), (0.1, 0.2), (0.3, 0.5), (0.5, 0.9), (0.8, 1.2), (1.0, 1.3), (1.2, 1.4), (1.4, 1.5), (1.5, \frac{\pi}{2}), (0.2, 1.5), (0.05, 0$ all negative, stand as a numerical cross-check, with Σ's own cell giving -0.7331 at $m = 128$ (converged value -0.7309566 by Theorem 73).*

Since Σ sits at $(\beta, \pi/2 - \beta)$ with $\beta < \pi/2 - \beta$, it lies in that family, and the class of competitors covered is strictly larger than $\mathcal{D}$ and is checkable from the sign pattern alone. (The restriction to *anchored* intervals is essential: Remark 26 exhibits $E_1 = E_2 = [0.4, 1.2]$, unanchored, with $\sup \frac{1}{2}\delta^2 Q / \|\eta\|^2_{L^2(0,\pi)} = +0.4123$.)

The constant in (44) is 99.9% of the sharp $c^* = 0.7309566\ldots$. What closed the earlier factor of 4.8 was giving up the pointwise splitting in two stages: Theorem 48 uses three separate Poincaré constants, of which $P_3 = 1$ is marginal and $P_2 = 1.5035$ nearly so; Theorem 72 replaces all three by two weighted eigenvalues that see the whole of each half at once; and Theorem 73 replaces those two by one system that sees both halves at once, losing nothing.

A correction this forced. Earlier drafts quoted the sharp constant as 0.7323, the P1 value at $m = 256$. Rayleigh–Ritz gives an upper bound on an eigenvalue and that sequence had not converged; it decreases through $0.731311, 0.731247, 0.731073$ at $m = 512, 1024, 2048$ towards $c^* = 0.7309566$. Every certificate in this note is a lower bound and every finite-element number an upper bound, so nothing crossed and no conclusion moves; the target was mis-stated by 0.2%.

Proposition 92 (H^1 stability)**.** *The second variation controls the derivative as well: measured against correctly assembled mass and stiffness matrices, the largest c_0 with $\frac{1}{2}\delta^2 Q[\eta] \leq -c_0 \|\eta\|^2_{L^2(0,\pi)} - c_1 \|\eta'\|^2_{L^2(0,\pi)}$ is*

c_1	0	0.1	0.3	0.5	0.7	1.0
c_0	0.7323	0.6114	0.3571	0.0760	-0.2560	-1.0000

at $m = 256$, each converging in m. So $\frac{1}{2}\delta^2 Q \le -0.357\|\eta\|^2_{L^2} - 0.3\|\eta'\|^2_{L^2}$, and the frontier is close to $c_0 = 0.732 - 1.31\,c_1$, vanishing near $c_1 = 0.56$.

Two simplifications are worth recording. First, the σ term has no singularity: since $-2\int \eta\eta' \tan t = -\int(\eta^2)'\tan t = \int \eta^2 \sec^2 t$ when $\eta(0) = \eta(\pi/2) = 0$, and $\sec^2 t - \tan^2 t = 1$,

$$-\int_0^{\pi/2} \big(\eta \tan t + \eta'\big)^2 dt = \int_0^{\pi/2} \big(\eta^2 - \eta'^2\big)dt,$$

checked to 10^{-12} on five test functions, so (24) may be written with bounded coefficients throughout. Second, the sharp L^2 constant is 0.7309566, not the 0.7285 of an earlier draft; that figure came from a mass matrix that gave the endpoint hat at $\theta = \pi$ the interior weight $2h/3$ instead of $h/3$.

Remark 93 (The size of $\mathcal{D}$, and what Theorem 71 is not)**.** It is not a proof that Σ is optimal, and the reason is quantitative.

Measuring the radius of $\mathcal{D}$ along $H_\Sigma + \varepsilon \sin(2k\theta)$ gives $\varepsilon_k = 0.0864,\ 0.0117,\ 0.0047,\ 0.0036,\ 0.0021$ at $k = 2, 4, 6, 8, 10$, so $\varepsilon_k k^2$ stays between 0.17 and 0.35: the radius scales like k^{-2}. $\mathcal{D}$ therefore contains a ball in the C^2 norm and in no weaker one, which is exactly what a bound on H'' should give. It is a genuine ball rather than a sliver, of 60 random eight-mode perturbations with coefficients of size $0.02/k^2$, 51 land but every member is a small C^2-perturbation of Σ. **Theorem 71 is thus a C^2-local statement**, with explicit convex hypotheses in place of a neighbourhood of unspecified size.

In every one of those probes the binding constraint was (b); condition (c) never became active. That is a statement about the probe and not about the geometry: (c) is *independent* of (b). Adding $\delta \sin(\theta - \pi/2)$ on $[\pi/2, \pi]$ and 0 on $[0, \pi/2]$ leaves $H + H''$ identically unchanged, so (b) and the gauge hold exactly, while the ceiling facet grows by δ and c_y grows by $\delta \sin t \cos t$; at $\delta = 0.2243\ldots$ the maximum M crosses $\frac{1}{2}$. So (c) cannot be derived and must remain a hypothesis.

$\mathcal{D}$ does not reach Gerver's cap, and by Remark 94 injectivity genuinely fails once the density exceeds 1. The failure is sharp in the condition but gentle in magnitude: comparing the flux $\frac{1}{2}\int(\alpha_2^+)^2$ with the area actually swept, the excess is $5.4\cdot 10^{-7}$ at density 1.047 and $7.4\cdot 10^{-5}$ at 1.267, and Gerver's own excess is about 0.25%. Widening $\mathcal{D}$ therefore does not need a weaker constant; it needs an upper bound on that excess which is *concave* in H, so that subtracting it preserves the concavity of Q. Truncating the sweep at $\inf_{t'} s_{22}(t, t')$ fails because that infimum is concave in H and $\frac{1}{2}\lambda^2$ then loses the convexity the Mamikon step supplies; the first-capture decomposition is exact but its parameter domain stops being of the form $0 \le s \le \alpha_2$, so its area is no longer quadratic in H. So the honest reading is that the one-corner architecture transfers to the ambidextrous problem on an explicit convex C^2-neighbourhood of Σ, and that enlarging that neighbourhood is where the remaining difficulty sits.

Remark 94 ((RC) is sharp)**.** The constant 1 in the forcing $R = 1 - (G + G'')$ of Theorem 40 is the corridor width: it enters through $c(t) = (F-1)\mu_t + (G-1)\nu_t$, so $\langle c(t), \nu_{t'}\rangle = n + G - 1$ and $n'' = -n + 1 - (G + G'')$. The argument therefore cannot tolerate a density above 1. Nor can the conclusion: perturbing Σ so that the maximum density passes through 1, the face-2 sweep acquires self-intersections exactly there. At densities $0.838, 0.921, 0.976$ there are none; at $1.004, 1.047, 1.119$ there are $160, 1510, 3086$ among $\binom{301}{2}$-many tested pairs, and a second mode gives 0 at 0.976 against 200 and 526 at 1.045 and 1.114. So (b) is sharp as stated.

22 Formalisation

Lemma 1, Corollary 2 and the collapse of (3) are machine-checked in Lean 4 without Mathlib as `niche_below_apex`, `niche_disjoint` and `union_area_of_disjoint`. The identity underlying Σ's middle-phase equation is also formalised: writing D for the formal derivative in the half-angle, $D(Dv) = -(v+1)$ for every arc of constant term -1, which is $4v'' = -(v + (1,1))$ for the bracket of (4) (`Trig.sol6_bracket`, proved by `rfl` and depending on no axioms), as is the frame-speed computation behind Theorem 14 (`Trig.sol1_speed_mu`) and the covering criterion of Theorem 11 (`strip_covers_iff`). `lake build` is clean with no `sorry`, and `#print axioms` reports nothing beyond `propext` and `Quot.sound`. Trigonometric quantities are carried as formal symbols with the sign hypotheses $\sin t \geq 0$, $\cos t \geq 0$, which is all Lemma 1 uses.

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